

Flood-v4.0 — Multi-Hazard Flood Screen: Technical Documentation

Open, validated, 30 m coastal + fluvial + pluvial design-event flood-depth model for four Southeast Asian cities (Singapore, Kuala Lumpur, Bangkok, Jakarta), present-day and four climate scenarios. This document describes the model as implemented; the companion manuscript (docs/paper/safe2026-v5.tex) gives the scientific framing and results.

New to flood modelling? Start with the **plain-language glossary in §11.2** — it defines every term used here (coastal/fluvial/pluvial, bathtub vs inertial, HAND, return period, the trust scores, ...) in everyday language. The parameter register in §11.4 opens with a plain-language reading guide, and each parameter group there starts with an "in plain terms" line, so a non-specialist can follow what every setting does without reading the equations.

1. Overview & scope

In plain terms. This document describes a computer model that draws flood maps. For each 30-metre square of ground in four cities it answers one question — how deep would the water be? — and it answers it three times over: once for the sea coming inland, once for rivers overflowing, and once for rain falling faster than the drains can take it. It then keeps the worst of the three. It repeats all of that for several severities of event and for two future climate pathways. Everything it uses is free and public, so anyone can rebuild it and check the answer — which is the point.

Purpose. Produce a license-clean, third-party-reproducible screen of physical flood hazard at a resolution (30 m) and hazard coverage (three perils) that no *open* product currently offers for these cities, so that exposure can be screened, priced, and disclosed by agencies that cannot license a commercial surface.

Coverage.

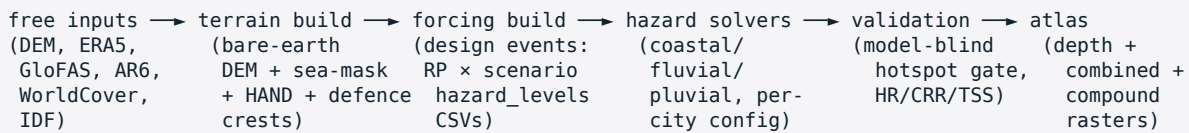
- **Cities:** Singapore, Kuala Lumpur (+ Port Klang), Bangkok, Jakarta.
- **Hazards:** coastal (tide + surge + sea-level rise), fluvial (river), pluvial (rain-driven / drainage-overwhelm).
- **Resolution:** 30 m, on a per-city UTM grid.
- **Return periods (RP):** 10, 100, 1000 year design events.
- **Scenarios:** present day (treated as SSP5-8.5 / 2020) plus four forward combinations — SSP2-4.5 and SSP5-8.5 × 2050 and 2100.
- **Primary output:** per-hazard flood **depth** (metres) rasters + a per-pixel **combined** (max-of-hazards) layer + a **compound** joint-exceedance layer.

Design philosophy. Every input is free and openly licensed; terrain is the dominant residual bias so the sole flood surface is an openly-constructed bare-earth DEM; and trust is established not by reproducibility alone but by a **pre-registered, model-blind location-skill gate** that scores documented flooded/dry localities under a discipline that forbids re-tuning.

Method principle — regime-matching, not one-size-fits-all. Solvers and referencing are matched to each city's physical flood regime rather than applied uniformly (see §6). The clearest example: the pluvial layer uses *fill-spill* (depression storage) on depression-dominated terrain and a *HAND-based handfill* (near-drainage rain pooling) on canal-dense cities — the choice is made on terrain physics, and adopting the wrong one measurably degrades the validation gate (documented in §6.3 and §10).

2. System architecture & pipeline

Data flow, left to right:



Repositories / directory layout (working tree D:\GPTs\Projects\flood-v4.0):

- dem/ — flood terrain. dem/<city>/ per-city conditioned bare-earth DEMs; dem/_hand/ derived Height-Above-Nearest-Drainage rasters; dem/_diag/ derived diagnostics including the seawall-raised DEMs. *Derived rasters under dem/_hand, dem/_diag, and sea_mask_*_framefix are git-ignored and regenerated by committed scripts.*
- data/<city>/ — per-city forcing (hazard_levels_<scenario>_rp<rp>.csv), sea-masks, river/drainage masks, runoff-coefficient rasters, and the validation manifest/ (hotspot registers).
- scripts/ — the model (run_multihazard.py), the gate (validate_hotspots.py), the terrain/sea-mask/HAND/defence construction (_fix_*.py, apply_pumped_polder.py), and visualisation (_viz_*.py).
- repro/ — resumable end-to-end drivers (run_atlas_fixed.sh, run_fixed_coastal_rp100.sh).
- outputs/ — model outputs (see §8).
- docs/ — this documentation, the paper, and the limitations register.

Two-interpreter rule. The flood model runs under a dedicated Python 3.14 interpreter (C:/Users/Daniel/AppData/Local/Python/pythoncore-3.14-64/python.exe), kept separate from the geospatial-preprocessing environment. All scripts/ and repro/ calls invoke that interpreter explicitly.

3. Input data & licensing

All inputs are free to use and redistribute, which is what makes the atlas license-clean and rebuildable from scratch by a third party.

Input	Role	Source / licence
Copernicus GLO-30 DEM	base elevation surface (pre-correction)	ESA, free
ICESat-2 (ATL08) + GEDI L2A	lidar ground truth for DEM calibration	NASA, open
ERA5-Land	meteorological reanalysis (rainfall context)	Copernicus C3S, free
GloFAS	fluvial discharge / river return levels	Copernicus EMS, free
IPCC AR6 sea-level projections	scenario sea-level rise by SSP × horizon	IPCC, open
ESA WorldCover	land-cover → runoff coefficient / impervious raster	ESA, CC-BY
National IDF standards	design rainfall the pluvial layer is anchored to	each national met service

Why national IDF matters. Global-synthetic design rainfall under-states the four national Intensity–Duration–Frequency standards by a large margin (a ~28–62 % deficit), so the pluvial forcing is anchored to each country's published IDF rather than to a global product. This is the single most important reason the pluvial layer is credible here.

4. Terrain: bare-earth DEM construction

In plain terms. Water runs downhill, so the ground-height map is the single most important input. The problem: satellites measure the height of whatever they see from above, which over a city means rooftops and treetops, not the ground. Buildings therefore look like dry hills, and streets that really flood get hidden underneath them. This section is how we strip the buildings and trees off to recover the real ground level, and how we check the result against laser height measurements taken from two space missions. We do the stripping separately for each city and never reuse one city's settings on another.

Rationale. Terrain is the dominant residual bias in a flood screen, and the raw Copernicus surface includes buildings and canopy. Existing bare-earth products either carry a non-commercial licence or

are too coarse, so the model builds its own.

Method. The Copernicus surface is corrected to ground by a **per-city building-and-canopy error model** calibrated against *held-out* ICESat-2 and GEDI lidar, roughly halving terrain error while remaining license-clean. The corrected, hydrologically-conditioned DEM is the sole flood surface.

4.1 The error model

The Copernicus GLO-30 surface z_{DSM} [R17] over-states ground by the building and canopy it contains:

$$z_{\text{DSM}}(x,y) \approx z_{\text{ground}}(x,y) + h_{\text{building}}(x,y) + f \cdot h_{\text{canopy}}(x,y)$$

z_{ground} (the wanted bare-earth) is recovered by subtracting the two bias terms and a flat offset.

Variables, with rationale and reference:

Symbol	Meaning	Value / source	Rationale	Reference
h_{building}	fractional-coverage building height	Google Open Buildings	per-pixel building bias; masked + back-filled from true-ground neighbours	Sirko et al. 2021 [R13]
h_{canopy}	canopy height (10 m)	ETH global CHM	the vegetation bias term	Lang et al. 2023 [R12]
f	canopy-penetration coefficient $\in [0,1]$	fit per city; 0.07-0.29	radar partially penetrates canopy, so only a fraction of h_{canopy} is bias; fitted not assumed	this work
b_0	flat true-ground offset	fit per city; +0.24 ... +1.46 m	residual datum/systematic offset after the two bias terms	this work
datum	vertical reference	EGM2008	the GLO-30 native datum; all WSEs/MSL offsets share it	—
subsidence δ	per-city present-day subsidence	published InSAR rates	applied to the present-day surface; projected-subsidence reported separately	Chaussard 2013, Abidin 2011, Phien-wej 2006 [R20-22]

Calibration. The two free parameters (b_0 , f) are fit **per city** against held-out **ICESat-2 ATL08** [R14] and **GEDI L2A** [R15] spaceborne-lidar ground returns, and are **never transferred between cities** — the discipline that makes "one method, four cities" honest rather than four bespoke tunings. Against held-out lidar the bare-earth roughly halves mean-absolute terrain error and removes the high bias (Bangkok MAE 1.43→0.78 m, bias +1.00→−0.14 m). The fit is product-invariant: re-deriving KL on a second canopy model moves b_0 by 3 mm. Rationale for *not* using FABDEM [R11] (the standard open bare-earth): its non-commercial licence cannot back an openly-redistributable atlas.

Per-city specialisation.

- **Bangkok / Jakarta / Kuala Lumpur:** the building-and-canopy error model is fit and applied directly; the result is the `_conditioned` / `_debiased` DEM.
- **Singapore (hybrid).** Spaceborne-lidar ground truth over dense, reclaimed Singapore is too noisy to calibrate the error model. Substituting the best independently-validated open bare-earth product (DeltaDTM [R28]) for the *low* terrain and compositing *high* ground from the surface model turns the weakest case into a validated one. Without this, the uncorrected product's coverage-capped hills collapse their height-above-drainage and spuriously flood the island's highest dry controls.

Coastal defences baked into the terrain. Continuous coastal/riverbank defences are sub-grid in the raw DEM, so a continuous-coastline crest is raised onto the DEM at each city's **citable, contemporaneous design crest** (above MSL):

City	Crest (m above MSL)	Provenance
Singapore	+3.0	PUB pre-2011 reclamation standard (the East Coast is pre-2011; +4 m is post-2011 only)
Bangkok	+2.10	BMA dike crest (UNESCAP)
Jakarta	+2.0	existing wall $\approx$ 1–2 m (Muara Baru 2 m); coastal here is subsidence-dominated, so crest barely changes the result

Built by `scripts/_fix_coastal_seawall.py` → `dem/_diag/<city>_seawall.tif`.

Sea-mask construction + frame-fix. The sea-mask is value **1 = land**, **0 = sea** (note: inverted from its name). The raw NaN-BFS sea-mask sweeps the DEM's nodata border frame (W/E/N edges) into "sea",

which the coastal solver then seeds from, producing spurious lateral-edge flooding.
 scripts/_fix_sea_mask_frame.py strips the frame by morphological opening and unions the real ≤ 0 m water back in (data/<city>/sea_mask*_framefix.tif). This matters specifically for the **inertial** coastal solver (frame seeds bypass attenuation), less so for a bathtub fill.

HAND rasters. Height-Above-Nearest-Drainage rasters (dem/_hand/) are derived from the conditioned DEM for the fluvial and pluvial referencing in §6. A **main-stem** HAND restricts "drainage" to channels above a flow-accumulation threshold (≥ 50 km² catchment), versus a **dense** HAND that treats the full canal/drain network as channel. Built by derive_drainage_mask_from_accumulation and the scripts/_fix*_mainstem_hand.py helpers.

5. Forcing & climate scenarios

In plain terms. "Forcing" is how much water the model is told to throw at the city: how high the sea rises, how much river flow arrives, how hard it rains. Those numbers come from tide-gauge records, river-flow archives and each country's own official design-rainfall standard. They are also the only thing that changes between scenarios — the model itself is byte-for-byte identical for the present day and for 2100, so any difference you see between two maps is the climate input, never a change of method.

A note on "return period": **RP100** does not mean "once every 100 years". It means a severity with a 1-in-100 chance of being exceeded in any given year. Two RP100 events can land in consecutive years without anything being wrong.

Design events. Each hazard is driven at three return periods (RP 10, 100, 1000). The forcing for a given (city, scenario, RP) cell is stored as data/<city>/hazard_levels_<scenario>_rp<rp>.csv — these are the committed, tracked inputs the atlas driver consumes.

Scenario stems (driver SCEN array):

Stem	Label	Horizon
ssp585_2020	present day (read as SSP5-8.5)	2020
ssp245_2050	SSP2-4.5	2050
ssp245_2100	SSP2-4.5	2100
ssp585_2050	SSP5-8.5	2050
ssp585_2100	SSP5-8.5	2100

→ 4 cities × 5 stems × 3 RPs = **60 atlas cells**.

Each (hazard, RP) row's water_level_m is built as below; the GEV/Gumbel maths live in scripts/gev_utils.py, the assembly in scripts/build_hazard_levels.py.

5.1 Coastal still-water level (water_level_m = peak WSE)

WSE_coastal(RP, scenario) = MSL_offset + surge_GEV(RP) + SLR_delta(scenario) (tide folded into the gauge baseline)			
Variable	Value / source	Rationale	Reference
WSE_GEV(RP)	GEV fit to UHSLC annual-maximum total still-water levels (tide + surge, de-meaned to local MSL — not de-tided ; the tide dominates and must be retained), ppf (1–1/RP)	annual-maxima GEV of total water level is the standard extreme-value basis for coastal design levels; consistent with global extreme-sea-level reanalyses	fetch_uhslc_gauge.py; Muis 2016/2020 [R9,R10]
GEV shape ξ	MLE, clamped to [–0.5, 0.30]	unclamped MLE allowed heavy Fréchet tails → unphysical RP1000; 0.30 validated vs PUB ponding 0.07–0.76 m	this work
MSL_offset	SG 1.1588 · JKT 0.9976 · BKK 1.1785 m	converts the EGM2008 DEM datum to local MSL; observed, not gate-tuned	derive_msl_egm2008_offsets.py
SLR_delta	AR6 tide-gauge zarr, p50, by SSP×horizon, added to baseline	scenario sea-level rise is an additive mean shift on the still-water level	IPCC AR6 [R6]

Present-day RP10→RP100 rises only ≈ 0.15 m (tide-dominated), so the coastal hazard is severe even at RP10 and grows slowly with RP.

5.2 Fluvial design stage (water_level_m = stage above channel bed)

GloFAS design **discharge** is converted to a **stage** via Manning's wide-rectangular inversion (gev_utils.mannings_stage):

$$Q \approx (1/n) \cdot w \cdot d^{(5/3)} \cdot \sqrt{S} \quad \Rightarrow \quad d = (Q \cdot n / (w \cdot \sqrt{S}))^{(3/5)}$$

Variable	Meaning	Source / rationale	Reference
Q	design peak discharge (m ³ s ⁻¹)	GloFAS v4 reanalysis design return levels	Alfieri 2018 [R30]
n	channel Manning roughness	per-city channel value	Manning [R26]
w	channel width (m)	per-city	—
S	channel slope (m/m)	per-city, from the DEM	—
d	stage above bed (the fluvial water_level_m)	mapped through HAND in §6.2	—

(w/d > 5 wide-channel assumption is checked and warned on violation.)

5.3 Pluvial net excess rain (water_level_m = excess depth, m)

Design rainfall comes from each **national IDF** standard, fit with a **two-anchor Gumbel** (_refit_pluvial_ifd.fit_gumbel); two (RP, depth) anchors *exactly* determine the two Gumbel parameters:

```
y_T = -ln(-ln(1 - 1/T));    σ = (x_b-x_a)/(y_b-y_a);    μ = x_a - σ·y_a
design_rain(RP) = μ + σ·y_RP
excess_depth_m = max(0, design_rain(RP) - drainage_loss) / 1000      (runoff coeff applied later, in-model)
```

Variable	Value (per city)	Rationale	Reference
IDF anchors (RP_a, x_a), (RP_b, x_b)	e.g. KL: (2 yr, 90 mm), (100 yr, 165 mm)	the published national IDF design-rainfall standard	PUB / JPS-MSMA / TMD-RID / BMKG [R23]
Gumbel ξ	0 (Gumbel, not GEV)	the standard form for IDF design rainfall; two anchors fix μ,σ exactly	—
drainage_loss	e.g. KL 70 mm	the drain-capacity threshold removed before ponding (drainage design level)	national drainage standard [R23]
runoff coeff C	raster (WorldCover); SG/KL 0.75, JKT 0.80	impervious fraction → runoff; applied once, in the model (verified, no double-count)	Zanaga 2022 [R16]

National IDF anchoring closes a **28-62 %** design-rainfall deficit that global-synthetic products carry over the tropics — the single most important reason the pluvial layer is credible here.

5.4 Climate scaling (future scenarios)

Hazard	Scaling	Rationale	Reference
Coastal	additive AR6 SLR delta (§5.1)	SLR is a mean-level shift	IPCC AR6 [R6]
Fluvial / Pluvial	multiplicative first-order Clausius-Clapeyron : factor = (1+cc·ΔT) ^h	warmer air holds ~cc more moisture per °C; uniform across RP (see note)	Lenderink 2017 [R5]

```
cc = 0.07 / °C      (Clausius-Clapeyron, sub-daily tropical rainfall)
h = 1.0 (pluvial, linear rain→stage) | 0.6 (fluvial, Manning Q^0.6)
ΔT by scenario:  SSP2-4.5/2050 ≈ 1.0 °C · SSP2-4.5/2100 ≈ 2.1 °C
                  SSP5-8.5/2050 ≈ 1.5 °C · SSP5-8.5/2100 ≈ 4.0 °C      (rel. 2020; AR6 SPM)
```

Important (implementation fact). build_hazard_levels._gev_cc_factor *perturbs the GEV by scaling both location and scale by α=(1+cc·ΔT)*. Because GEV is a location-scale family, the quantile ratio is **exactly α for every return period** — a uniform first-order CC intensification, **not** return-period-specific or super-CC. (The paper §3(iv) and the code docstring were corrected to say so, 2026-06-22.)

6. Hazard models

All three hazards are produced by scripts/run_multihazard.py, which dispatches to solver modules under model/: inertial_wave_model.py (coastal), flood_depth_model.py (HAND/bathtub/connectivity), pluvial_model.py (fill-spill cascade), and pluvial_rain_model.py (rain-on-grid). The per-city invocation is encoded in repro/run_atlas_fixed.sh. Outputs are written per hazard under

<out>/<hazard>/rp_<rp>/<hazard>_depth_<scenario>_<horizon>_rp<rp>.tif.

6.0 Common notation and forcing decomposition

Throughout this section (for everyday definitions of Manning's n , WSE, HAND, and the rest, see the plain-language glossary, §11.2):

Symbol	Meaning	Units
$z(x, y)$	bed elevation (the bare-earth flood DEM), NaN = nodata/wall	m (EGM2008)
$d(x, y, t)$	water depth above bed	m
$\eta = z + d$	water-surface elevation (WSE)	m
q	unit discharge across a cell interface (depth-integrated velocity, $q = u \cdot h$)	$\text{m}^2 \text{s}^{-1}$
h_f	flow depth at an interface = $\max(\eta_L, \eta_R) - \max(z_L, z_R)$	m
n	Manning's roughness coefficient	$\text{s m}^{(-1/3)}$
g	gravitational acceleration = 9.806	m s^{-2}
$\text{HAND}(x, y)$	height above nearest drainage	m
$\Delta x, \Delta y, \Delta t$	grid spacing (30 m) and timestep	m, s

One forcing field, three interpretations. Each (hazard, RP) row of the hazard_levels CSV carries a single water_level_m, but its meaning differs per hazard — this is the central wiring fact of the model:

- **Coastal** — water_level_m is the **peak still-water level** (absolute WSE, in the DEM datum) = MSL + tide + surge + SLR. It is the *boundary condition*, not the answer; depth is solved.
- **Fluvial** — water_level_m is the **river stage at the return period**, above which the *overbank* excess $\max(0, \text{stage}_{\text{RP}} - \text{bankfull})$ is mapped through HAND.
- **Pluvial** — water_level_m is the **net excess rain depth** ($\text{design}_{\text{rain}} - \text{drainage}_{\text{loss}}$), later multiplied by a per-cell runoff coefficient to become a runoff volume.

The **runoff coefficient** $C(x, y)$ (from WorldCover land cover, --runoff-coeff-raster, scalar fallback --runoff-coeff) scales rain to runoff: per-cell runoff volume = excess $\times C \times \text{cell_area}$.

6.1 Coastal — local-inertia shallow water vs bathtub

In plain terms. *There are two ways to work out how far the sea gets inland. The quick way — the "bathtub" — fills every low spot connected to the sea, instantly, as though the city were a bath. It has no notion of momentum, friction, or time, so it floods ground fifteen kilometres inland at the same moment as the shoreline. The slower way, which we use, steps the water forward in time with momentum and friction, so land the water cannot physically reach during a six-hour surge stays dry.*

This matters more than it sounds. Behind a working sea defence the two answers differ by about five times, because the bathtub leaks through canals that a 30-metre map cannot resolve and quietly floods the whole protected interior. Every headline coastal number in this atlas is read off the slower, physical solver.

This is the headline methodological choice. The atlas computes coastal inundation with a **2-D local-inertia shallow-water solver** (model/inertial_wave_model.py, --coastal-solver inertial), and explicitly rejects the **bathtub** (connectivity) fill that open tools default to. The two are derived below, with the reason the bathtub over-predicts.

6.1.1 The shallow-water equations and the local-inertia simplification

The full 2-D shallow-water (Saint-Venant) momentum equation for unit discharge q in the x-direction is

$$\frac{\partial q}{\partial t} + \frac{\partial (q^2/h)}{\partial x} + g \cdot h \cdot \frac{\partial \eta}{\partial x} + g \cdot n^2 \cdot q \cdot |q| / h^{(7/3)} = 0$$

(local accel) (advection) (pressure grad) (friction)

For slowly-evolving flood waves at **low Froude number** ($Fr = u/\sqrt{gh} \ll 1$ — true for coastal/urban flatland inundation), the nonlinear **advection** term $\partial (q^2/h)/\partial x$ is negligible and is dropped. Discretising the remaining three terms explicitly in time and solving for q^{n+1} gives the **Bates, Horritt & Fewtrell (2010)** update used here:

$$q^{n+1} = \frac{q^n - g \cdot h_f \cdot (\partial\eta/\partial x) \cdot \Delta t}{1 + g \cdot n^2 \cdot |q^n| \cdot \Delta t / h_f^{(7/3)}}$$

where $h_f = \max(\eta_L, \eta_R) - \max(z_L, z_R)$ is the interface flow depth (the "two-cell" depth that lets water cross a sill only when the higher WSE exceeds the higher bed). The numerator advances momentum under the **water-surface-slope pressure gradient**; the denominator applies **Manning friction** semi-implicitly (it appears as q^{n+1} in friction but $q^n \cdot |q^n|$ in the quadratic, which keeps the scheme stable without a matrix solve). Mass is then conserved by the explicit **continuity** update:

$$d^{n+1}_{\{i,j\}} = d^n_{\{i,j\}} - (\Delta t/\Delta x)(q_{x,\{i,j\}} - q_{x,\{i,j-1\}}) - (\Delta t/\Delta y)(q_{y,\{i,j\}} - q_{y,\{i-1,j\}}), \text{ clipped } \geq 0$$

Stability (CFL). The explicit scheme requires $\Delta t \leq \alpha \cdot \min(\Delta x, \Delta y) / (\sqrt{g \cdot d} + |u|)$ with safety factor $\alpha = \text{CFL_ALPHA} = 0.7$. The solver recomputes Δt adaptively every step (`_adaptive_dt`) from the current max gravity-wave celerity $\sqrt{g \cdot d}$ and max interface velocity, capped at `--inertial-dt-max` (default 30 s). Sea (Dirichlet) cells are excluded from the CFL test so their large depths don't throttle the land timestep.

Implementation. Hot kernels (`_flux_x/_flux_y/_continuity`) are JIT-compiled with numba (`@njit(parallel=True, fastmath=True)`) when available, with verified pure-numpy fallbacks. The domain is auto-cropped to the bounding box of cells that can ever wet ($\text{sea} \cup \{z \leq \text{peak_WL}\} + 32\text{-cell pad}$), often halving per-step cost. $g = 9.806$, $\text{MIN_DEPTH} = 1$ mm (interfaces shallower than this carry no flux).

6.1.2 Boundary condition — surge hydrograph on an SLR floor

Sea cells are held at a **time-varying Dirichlet WSE** $wl_fn(t)$ (`_surge_hydrograph`), a synthetic asymmetric triangular surge riding on a **permanent still-water floor**:

```
floor = MSL_EGM2008 + SLR          (this row's coastal_delta_m)
0 → 3h : linear ramp 0 → peak_WSE (gentle start; a dry-bed step shocks
                                   the explicit scheme)
3h → 4h : hold at peak_WSE
4h → 6h : linear recession peak_WSE → floor (NOT → 0)
```

The recession stops at the floor, **not zero**, so sea-connected land below MSL+SLR stays inundated after the surge withdraws — the physically correct persistent SLR signal. When $\text{floor} > 0$ the early-convergence test is disabled (`convergence_window` set huge) so the solver actually simulates the full ramp/hold/recession rather than exiting on the sub-tolerance warm-start state.

Warm-start. Two stacked initialisers cut steps-to-solution 3–5×: (1) `subsea_init` pre-floods below-MSL land to the floor (steady-state start), and (2) the **previous RP's** peak depth seeds the next RP so only the surge *increment* must be resolved. RPs are therefore run in ascending order with depth chained forward.

Post-hoc physical cap. Narrow inlets/steep gradients can numerically amplify a few cells during the sustained peak hold. Depth is capped at

$$d \leq (\text{peak_WSE} - \text{bed}) + 0.2 \text{ m} \quad (2026-06-24: \text{bed}, \text{NOT } \max(0, \text{bed}))$$

the static maximum plus a velocity-head margin ($\approx \frac{1}{2}v^2/g$ for $v = 2 \text{ m s}^{-1}$) — removing artefacts without touching genuine depths or bulk extent. **The bed (not $\max(0, \text{bed})$) form is essential on subsided coasts:** clamping the bed at 0 caps below-MSL cells at the water level itself and silently under-states the water column in Jakarta's below-sea-level polders (the cap fix of 2026-06-24; see §10 and the depth caveat in §6.1.3). Note the resulting depths are *equilibrium-fill* values — the full column from surge surface to bed — which is the damage-relevant depth where a polder is sea-connected but an upper bound where active pumping holds it lower. The inertial solver resolves connectivity intrinsically, so **no BFS connectivity filter** is applied to its output (unlike bathtub).

6.1.3 The bathtub method, and why it over-predicts

The bathtub depth (`flood_depth_bathtub`) is purely static:

$$d(x, y) = \max(0, \text{WL} - z(x, y))$$

followed by a **connectivity filter** (`connected_flood_mask`): a 4/8-connected BFS from the sea boundary (and tidal-channel seeds) keeps only flooded cells hydraulically connected to open water, discarding isolated inland lows. There is **no momentum, no friction, and no time** — every cell connected to the sea and below WL floods to the full WL – z, regardless of how far inland it is or what gradient/roughness the water would have to traverse to get there.

This is the over-prediction mechanism, and it is an artifact of **solver architecture, not 30 m resolution**: on a gradient- and defence-rich delta, a momentum-and-friction solver leaves far-field low ground dry within a 6-hour surge because the water cannot physically arrive and persist, whereas the bathtub floods it instantly. The gap is **regime-dependent**, up to $\approx 5\times$ on the Bangkok delta. Every headline coastal exposure figure is read off the **inertial** layer.

	bathtub	local-inertia (this model)
Pressure gradient (backwater)	$\times$ static flat WSE	$\checkmark$ $g \cdot h \cdot \partial \eta / \partial x$
Mass conservation	$\times$ none	$\checkmark$ strict continuity
Friction / travel time	$\times$ instantaneous	$\checkmark$ Manning n, finite surge window
Connectivity	BFS post-filter (4/8)	intrinsic to the 2-D solve
Velocity / $h \cdot v$ hazard	$\times$	$\checkmark$ (optional)
RP100 bias on defended delta	up to $\sim 5\times$ high	reference
Cost	seconds	~ 30 min/RP/city (inertial)

--coastal-solver bathtub is retained as a fast screening fallback (and is used when no sea-mask is available); it is **not** used for the published atlas.

6.1.4 Datum, negative land, defences

- **Datum.** WSEs are referenced to local MSL via per-city EGM2008 offsets (--coastal-msl-egm2008): Singapore 1.1588, Jakarta 0.9976, Bangkok 1.1785.
- **Negative land.** Jakarta/Bangkok pass --no-clamp-negative-land: genuine below-MSL land exists (Jakarta subsidence) and clamping would erase real exposure.
- **Defences.** Continuous coastal crests are baked into the DEM (§4); Bangkok adds the pumped-polder post-process (§6.1.5).

6.1.5 Pumped polders (Bangkok only)

Bangkok's King's-Dyke CBD stays dry behind pumped polders, so `scripts/apply_pumped_polder.py` dries documented polder rings in post-processing (<out>_polder — the scored/published composite). Tested for Jakarta and **rejected**: Jakarta's Pluit/Muara Baru polders genuinely flood at RP100 and the flooded dry control there (Cipete) is fluvial, not coastal — so the polder fix is Bangkok-only.

6.2 Fluvial — HAND referencing vs other methods

In plain terms. River flooding is handled by asking, for every point in the city, "how far above the river am I?" — then raising the river by the design amount and seeing what ends up underwater. The subtlety is which river. A city is threaded with hundreds of small drains as well as a few big rivers, and the flow figure we have describes only the big ones. Measure height above every little drain and you flood the whole city; measure height above the main trunk the flow actually belongs to and you get something sensible. That choice is the substance of this section.

6.2.1 What HAND is

HAND (Height Above Nearest Drainage; Rennó et al. 2008, Nobre et al. 2011) is a terrain transform: every cell's vertical height above the channel cell its D8 flow path drains to. It renormalises absolute elevation into *elevation relative to the local river*, which is the natural vertical reference for river flooding. It is computed once per city from the conditioned DEM: D8 flow directions on the pit-/flat-resolved surface, a drainage network defined by flow-accumulation threshold, and for each cell the drop to its nearest network cell along the flow path.

6.2.2 Depth mapping (the model)

Fluvial depth is a **static stage-minus-HAND** map (`flood_depth_hand`), but the stage is the **overbank excess above bankfull**, not the raw return-period stage:

```
overbank = max(0, stage_RP - bankfull_stage)      (--fluvial-bankfull-rp sets bankfull; 0 here)
d(x,y)   = max(0, overbank - HAND(x,y))
d(channel) = 0                                     (burned channel cells masked – see below)
```

Cells with `HAND < overbank` flood, to a depth that is greatest at the channel (`HAND ≈ 0`) and tapers to zero at the inundation edge (`HAND = overbank`). Subtracting bankfull first means a return period only inundates the floodplain by the excess over the channel's design conveyance — below bankfull, the river is in-bank and the floodplain is dry.

Channel masking. `HAND` is 0 on the burned channel network, so every channel cell would "flood" on any overbank — but a canal carrying water is *conveyance*, not inundation, and many channels are below-grade (engineered beds), which would make the map appear to flood "underground". Channel cells are therefore zeroed (`depth = where(river_mask, 0, depth)`), mirroring the pluvial treatment of channels as sinks.

6.2.3 Main-stem vs dense-network drainage — the threshold that matters

The single most consequential fluvial parameter is the **flow-accumulation threshold** that defines "drainage" (`derive_drainage_mask_from_accumulation`):

- **Dense HAND** — the full canal/drain network is "channel". On canal-dense cities this labels a huge fraction of land as `HAND ≈ 0` (Singapore's dense hybrid `HAND`: 96.5 km² ≈ 19 % of land), so a modest stage floods ~40 % of the island and the *fluvial* layer silently **absorbs the pluvial hotspots** — the failure that crashed Singapore's gate from 0.84 to 0.13 when fluvial was reduced naively.
- **Main-stem HAND** — only channels draining above a **per-city accumulation threshold** count: **≥50 km²** for Singapore and Jakarta, **≥180 km²** (the larger Klang trunk) for Kuala Lumpur. The stage then floods land within reach of the *trunk* river, not of every minor drain. This corrects the fluvial extent (SG fluvial 180→28 km², channel 96.5→2.8 km²) and forces the pluvial hazard onto the pluvial layer where it belongs.

Per-city `HAND` rasters: Singapore `singapore_hand_v3_mainstem`, Jakarta `jakarta_hand_v3_mainstem`, Bangkok `hand_trunk_v3_debiased`, KL `kl_hand_mainstem_v3eth` (already main-stem at ≈0.9 % channel).

Provenance and reproducibility (all four now resolved; 2026-07-18). Every raster in the repo is git-ignored by convention; the *primary* inputs (bare-earth DEMs, masks, forcing) ship in the Zenodo archive's `inputs/`, and committed scripts regenerate the *derived* ones. The four fluvial `HAND`s sit in two categories:

- **Script-derived** (regenerated by the atlas driver from committed inputs): Singapore `scripts/_fix_sg_mainstem_hand.py`, Jakarta `_fix_jakarta_mainstem_hand.py`, and — as of 2026-07-18 — KL `scripts/_fix_kl_trunk_hand.py`. A plain flow-accumulation threshold does **not** reproduce the KL trunk (a **≥180 km²** threshold gives only ~3 km² of channel and IoU 0.82 against the shipped `HAND` — the shipped Klang trunk is a denser v2-lineage network), so the topology is shipped as an explicit **frozen drainage mask** (`data/kuala_lumpur/trunk_drainage_mask_utm47n.tif`, 12,362 channel cells) and the builder computes `compute_hand(DEM, mask)` from it. Verified: **regenerates the shipped raster to max|Δ| = 0 (bit-identical)**; the mask now ships in the KL Zenodo inputs.
- **Primary frozen input** (shipped in the archive, not script-derived): Bangkok `hand_trunk_v3_debiased`. Its drainage is the v2 Chao-Phraya-trunk network **plus an opaque debiasing step** that no committed script reverses (from-mask and from-accumulation rebuilds reach only IoU ≈ 0.65 and would change the shipped fluvial layer). It is therefore shipped as a primary frozen input in the Bangkok Zenodo `inputs/` — **the same category as the bare-earth DEMs**, which likewise have no in-repo builder and ship as frozen inputs. This is a documented, bounded provenance limitation, not an availability gap: the exact raster is obtainable and the Bangkok fluvial result reproduces from it.

Flat-delta note. *Even a main-stem HAND keeps a large fluvial footprint on a low-gradient delta, because everything sits within the trunk stage of the river (Jakarta fluvial ≈181 km²). This is physical, not a bug — the whole delta is the Ciliwung floodplain.*

6.2.4 HAND vs the alternatives

Method	Physics	Cost	Why not (for this screen)
HAND (used)	static stage on drainage-relative terrain	~seconds	the screening sweet spot — topographically coherent, no routing, scales to 60 cells
Bathtub fluvial	$\max(0, \text{stage} - z) + \text{BFS from channel}$	seconds	references absolute elevation, not the river; floods any low ground at $z < \text{stage}$ even if it is nowhere near a river
2-D hydrodynamic (inertial, array BC)	full local-inertia with a per-cell channel WSE field held as Dirichlet (the solver <i>supports</i> this — <code>wl_boundary</code> as an ndarray) and spill routed onto the floodplain	~30 min/RP	overkill for design-event <i>screening</i> ; needs reach-by-reach channel geometry/discharge the open forcing doesn't resolve at 30 m
Full hydraulic model (HEC-RAS etc.)	1-D/2-D coupled, surveyed cross-sections	hours-days	not license-clean, not rebuildable from free data, not scalable to the atlas

HAND is chosen because it captures the dominant control on fluvial extent — *height above the river* — at screening cost and from free terrain, while the heavier options need data the open pipeline cannot supply uniformly across four countries. The inertial *riverine* path exists in the code (array boundary condition) for future per-reach work.

6.3 Pluvial — four models, regime-matched

In plain terms. *This is flooding caused purely by rain arriving faster than the ground and the drains can carry it away — the "flash flood" that global flood products usually leave out entirely, and the biggest single hazard in three of our four cities. Rain floods a city in one of two ways, and which one depends on the shape of the land. On flat ground with lots of hollows, water collects in the hollows and spills from one to the next; that is Bangkok. In cities threaded with canals and open drains, water backs up along the drainage network and floods the low land beside it; that is Kuala Lumpur, Singapore and Jakarta. We use a different method for each, because using the wrong one measurably lowers the trust score in §7.*

Pluvial (rain-driven) flooding is where method choice matters most, because the physical mechanism differs by terrain: water either **collects in closed basins** (depression storage) or **backs up beside overwhelmed drains** (drainage-overwhelm). `run_multihazard.py` implements four `--pluvial-model` options. Two are in production use — `fillspill` and `handfill` — and two are alternatives (`raingrid`, `legacy`). All share the same forcing: `water_level_m` is the net excess rain depth, scaled by the per-cell runoff coefficient `C`.

△ **The atlas is built in two stages.** §6.3.1–6.3.5 below document the solvers and the original regime assignment (`fill-spill`: Bangkok, KL; `handfill`: Singapore, Jakarta), which is what the **Stage-1 solve loop** in `repro/run_atlas_fixed.sh` invokes. That Stage-1 atlas is then modified by the two applies below, and the outputs/paper reflect the post-retrofit state. Since 2026-07-13 both run inside the same driver as **STAGE 2**, so one command reproduces the shipped product:

1. **Canal-HAND retrofit (2026-07-04)** — a HAND `handfill` referenced to the real mapped drain network (not a synthetic dense net), applied under a selection rule rather than blanket-applied. Measured verdicts: **KL applied** (unioned per-pixel with its depression cascade), **Jakarta applied**, **Singapore rejected** (its shipped pluvial already scores strongly), **Bangkok rejected** (pumped flat delta — CRR collapses). The rule: retrofit only where the shipped pluvial-only HR is demonstrably low **and** the register's dry controls sit topographically above its positives; on flat pumped deltas engineered drainage decouples flooding from topography, so no HAND variant can discriminate. Run docs:

`docs/runs/2026-07-04-*canal-hand-pluvial.md`. 2. **Pluvial closing (2026-07-07)** — see `docs/runs/`.

§6.3.5 documents both stages and states the shipped post-retrofit configuration — read it for what the atlas and paper actually report. Gate numbers quoted inside §6.3.1–6.3.4 are **Stage-1 (pre-retrofit)** values, retained because they are the evidence that selected each city's base solver.

6.3.1 `fillspill` — catchment-routed fill-and-spill cascade

The production depression-storage model (`model/pluvial_model.py`) is a topologically-ordered **fill-and-spill cascade** — a screening-grade approximation of Barnes et al. (2020) Fill-Spill-Merge —

not a lumped fill. It is built once per city (`build_pluvial_topography`) and routed per RP (`route_pluvial_rp`).

Step 1 — depression inventory. Pit-fill the DEM (filled), then label connected components (8-connectivity) of filled – dem > 0. Each component is a depression, kept only if it passes three physical filters that screen out DEM noise and non-ponding features:

```
depth    d(x,y) = filled(x,y) - dem(x,y)
keep depression  iff  0.5 m ≤ max(d) ≤ 3.0 m    and    area ≥ 9 cells (≈0.8 ha)
```

(< 0.5 m = noise; > 3.0 m = quarry/valley/reservoir/artefact, excluded like sea; < 9 cells = sub-pixel inter-building voids in the GLO-30 DSM.)

Step 2 — hypsometric curve. Each depression stores its cell bed elevations sorted ascending. Its **volume-level** relation is

```
V(h) = cell_area · Σ_i max(0, h - bed_i)    (capacity = V(pour_elev))
```

inverted analytically (`_fill_level`) to find the water level *h* that holds a given inflow volume.

Step 3 — catchment supply. Per-cell runoff volume excess · *C* · cell_area is routed by **D8 steepest descent** on the raw DEM; every cell is credited to the depression its flow path terminates in (`compute_catchment_supply`, via terminal-label propagation in ascending-elevation order). Runoff that terminates in an excluded/edge sink is dropped (it drains away, exactly as sea/river sinks do).

Step 4 — spill cascade. Each depression's overflow destination is found by walking the *conditioned* D8 field until it reaches another depression or a sea/river sink (`build_spill_graph`). Depressions are then filled in **topological order** (`run_cascade`): if total inflow ≥ capacity, the depression fills to its pour elevation and the surplus cascades to its downstream destination; otherwise it fills partially via the inverted hypsometric curve. Processing downstream-after-upstream guarantees each depression's inflow is final before it fills.

Step 5 — paint depth. *d* = max(0, level_d – bed) for each depression's cells; cells below `WET_THRESHOLD_M` = 0.05 m are set dry.

This is **RP-dependent** (extent grows with return period) and **parameter-light** — no free stage parameter, just the runoff excess. It is physically correct exactly where flooding is basin storage: **flat depression-dominated deltas** (Bangkok) and any terrain whose lows are genuine closed basins.

6.3.2 handfill — rain pooling near open drainage

On canal-dense cities most excess rain never enters a closed basin — it sheets down open gradients and **backs up at low points beside saturated drains**. `fillspill` structurally reports almost no ponding there (Singapore: 1/38 hotspots). `handfill` models this drainage-overwhelm regime as a **HAND stage on the dense network**:

```
d(x,y) = max(0, stage - HAND_dense(x,y))    (flood_depth_hand on the dense HAND)
d = 0   where river_mask OR HAND_dense < 0.05 m    (channels = conveyance, not flood)
d = min(d, depth_cap)    (life-safety flash-flood bound)
```

It reuses the HAND depth kernel (§6.2.2) but on the **dense** HAND (the full drain network, not the trunk), so it floods the low land *adjacent to every open drain* to a rain-driven stage. Two parameters:

--pluvial-hand-stage sets the flood-prone **extent**, --pluvial-depth-cap bounds the **depth** (without the cap, stage – HAND over-deepens right next to drains). Production values: Singapore (hybrid HAND, stage 1.5 m, cap 0.5 m), Jakarta (`jakarta_hand_v3`, stage 2.5 m, cap 1.0 m). Result (*Stage-1, pre-retrofit*): Singapore pluvial 15→95 km² (present-day), gate HR 0.13→0.71 PASS; Jakarta TSS 0.66→0.76 PASS. Jakarta's base layer was subsequently unioned with a canal-HAND layer (Stage 2, §6.3.5), lifting its gate to 0.85/0.92/**0.77**; Singapore's base layer ships unchanged.

Return-period / climate scaling (2026-06-22). With a *fixed* stage the handfill layer is **forcing-invariant** — it ignores `water_level_m`, so the SG/Jakarta pluvial extent was identical at every RP and every scenario (SG 95.1 km², Jakarta 168.0 km² everywhere). The opt-in flag --pluvial-hand-stage-baseline <excess_m> fixes this: the effective stage scales with this row's rain excess, `stage_eff` = stage × clip(`level_m` / baseline, 0.4, 2.5), where baseline is the present-day

RP100 excess the stage was calibrated against (SG 0.0696, Jakarta 0.1300). At `level_m == baseline` the stage is unchanged, so the **present-day RP100 gate is preserved by construction**; other RPs/scenarios now respond (SG 49→95→129; Jakarta 108→168→217 km² at RP10/100/1000, pre-closing). **The shipped atlas carries the scaled layer** (measured: SG pluvial 56.7/111.2/150.2 km² at RP10/100/1000 present-day, ≥0.10 m) — historically applied post-rebuild by `repro/rerun_handfill_scaled.sh`, and since 2026-07-17 produced directly by the driver, whose SG/Jakarta lines pass `--pluvial-hand-stage-baseline (0.069602 / 0.129999)` in Stage 1.

Physical bracket on the base stages (anchored 2026-07-18; review C7 follow-up #3). The base stages were chosen with gate visibility (§10 item 3); the external standards do not *pin* either value, but they **bracket** it, which bounds the choice physically:

- **Singapore 1.5 m.** The dense-HAND reference surface sits at the DEM's drain elevation (engineered beds read below grade — limitations register #1), so the stage is the height of ponding *above the drain bed*: drain depth + surcharge above bank. PUB's Code of Practice on Surface Water Drainage requires roadside drains ≥ 0.6 m deep, classes ordinary open drains by ≤ 1 m / > 1 m depth (i.e. metre-deep drains are routine), sets freeboard at **15 % of drain depth** (CoP §7.3.4), and designs minor catchments (< 100 ha) to a 10-yr storm (CoP §7.1.3) — beyond which the drain surcharges. A 1.0–1.2 m outlet drain surcharged to bank-top plus ponding within PUB's own documented 0.07–0.76 m range spans ≈ **1.1–2.0 m above the bed**; 1.5 m sits mid-band.
- **Jakarta 2.5 m.** BNPB's reporting for the January-2020 monsoon event — the design-class event for this layer — recorded 268 flooded locations at depths **0.2–2.5 m**, six kampungs ≥ 2 m (Cipinang Melayu peaking ≈ 4 m). The 2.5 m stage is the top of BNPB's city-wide band: the layer floods land up to, and not beyond, the documented envelope of the design-class event. (Indonesian practice — SNI 03-2406-1991 / Pd T-02-2006-B — sets drain freeboard at 5–30 % of flow depth, same order as PUB's 15 %.)
- **KL / Jakarta canal-HAND stage_ref (0.5 / 0.3 m, §6.3.5).** MSMA (2nd ed., ch. 26) dimensions lined open drains at 0.5–1.2 m with 50–300 mm freeboard; the 0.5 m and 0.3 m reference stages are the depth class of the *mapped minor-drain network* they are referenced to, not the metre-deep outlet class above.

The honest provenance statement is therefore **GATE-SELECTED within a standards-bracketed range**: the bracket comes from PUB/MSMA/SNI + the BNPB event record; the specific value inside the bracket was chosen with gate visibility. §11.4 carries this classing.

6.3.3 raingrid — 2-D rain-on-grid (alternative)

The most physical pluvial option (`model/pluvial_rain_model.py`): the **same local-inertia solver as the coastal model** (§6.1.1), but with a distributed **rainfall source term** instead of a sea boundary:

<code>net rain rate = (excess · C) / storm_duration</code>	applied to land cells while <code>t < storm</code>
<code>d^{n+1} += rain_rate · Δt</code>	(source, on top of the SWE update)
outlets (sea u open channels): <code>d = 0</code> each step	(free-drainage sinks)

Manning's *n* is spatially variable (land-cover-derived, ≈ `clip(0.11 - 0.08·C, 0.03, 0.10)`). Drains can be made **finite-capacity** (`--drain-conveyance-m-s`): a perfect-sink mask (sea + major rivers) drains freely while minor channels shed only conveyance · Δt per step and pond the remainder when overwhelmed. Output is the **peak** depth over a storm-plus-settling window (default 1 h storm + 0.5 h settle), denoised (`denoise_min_cluster`, drop < 6-cell ≈ 0.5 ha speckle) and floored at 0.05 m. RPs are solved in a process pool (`--raingrid-workers`). *Not* in production: on steep terrain it over-drains (KL → ~0 pluvial), and where it works it is far costlier than handfill for a similar result.

6.3.4 legacy — lumped depression fill (deprecated)

The original model (`flood_depth_pluvial_ponding`): `d = min(WL, filled - dem)`. RP-frozen extent (the depression footprint is identical at every RP, only depth scales), superseded by `fillspill`. Retained for reproducing earlier results.

6.3.5 Model selection — two gate-arbitrated decisions

The shipped pluvial layer is the result of **two** decisions, taken three months apart and both arbitrated by the model-blind register rather than by preference. Stage 1 chose the *base solver* per city (2026-06); Stage 2 retrofitted a **canal-HAND** layer onto two of the four (2026-07-04). Both are documented here because the second changes what several earlier sections imply.

Stage 1 — fillspill vs handfill (the base layer)

Regime-matching is **empirically validated, not stylistic**: forcing the wrong model measurably degrades the model-blind gate (expanded register, present-day RP100). These are the **pre-retrofit** scores that selected each city's base solver:

City	Regime	fill-spill TSS [95% CI]	handfill TSS [95% CI]	base solver
Singapore	canal-dense island	fails (1/38 hotspots)	0.61 [0.41, 0.79] PASS	handfill
Jakarta	canal-dense delta	0.66	0.76 [0.61, 0.91] PASS	handfill
Kuala Lumpur	steep, sub-grid	0.65 [0.48, 0.81]	0.51 [0.23, 0.76]	fill-spill
Bangkok	flat khlong delta	0.34 [0.19, 0.50]	0.37 [-0.01, 0.72]	fill-spill

The two failure modes are instructive and **opposite**:

- **Bangkok (handfill fails).** On a flat khlong-laced delta *everything* is within 1.5 m of a drain, so handfill floods ≈ 617 km² (vs fill-spill's 179) — most of the delta. HR rises spuriously (flood-everything) but specificity collapses and the **TSS CI crosses zero** — it loses statistical significance. Fill-spill's depression geometry preserves specificity (CRR 1.00).
- **KL (handfill underperforms).** Handfill gives a more *coherent-looking* map (3.5k zones vs fill-spill's 12k tiny pits) and slightly higher HR, but floods two dry controls, dropping TSS 0.65→0.51. The spotty fill-spill is gate-optimal precisely because its scattered pits dodge the dry controls. KL's flash floods are fundamentally **sub-grid** at 30 m; neither model is ideal, and fill-spill is retained for the gate.
- **Singapore / Jakarta (handfill wins decisively).** Canal-dense, bounded terrain where the hotspots sit beside open drains — exactly handfill's regime — so it lifts both cities over the PASS bar where fill-spill cannot represent the mechanism at all.

Stage-1 decision rule. Use fillspill for depression-dominated terrain (closed basins: flat deltas, steep pits) and handfill for canal-dense cities whose flooding is drainage-overwhelm beside open drains. See §10 for the open caveat that handfill's stage was effectively gate-selected and needs re-anchoring to a documented value.

Stage 2 — the canal-HAND retrofit (2026-07-04)

Stage 1's handfill references a **synthetic dense HAND**. Stage 2 asks a sharper question: reference the stage to the **real, mapped drain network** instead. The retrofit adds a canal-HAND layer **unioned per-pixel** with the city's existing (Stage-1) layer, so it can only *add* wet cells:

```
pluvial_new(x,y) = max( pluvial_shipped(x,y), clip(stage_cell - HAND_canal(x,y), 0, cap) )
d = 0 on channels                                     (conveyance, as everywhere else)
HAND_canal      = height above the MAPPED drain network (OSM / river mask)
stage_cell      = stage_ref × clip(cell excess / baseline_excess, lo, hi)
```

City	HAND_canal network	stage_cell	cap
Kuala Lumpur	drainage_waterways_utm47n, pruned to drains with ≥ 0.5 km² flow accumulation	0.5 m × (excess/0.095), clip [0.4, 2.5]	3.0 m
Jakarta	river_mask_utm48s, unpruned (full network)	0.3 m × (excess/0.130), clip [0.15, 1.5]	3.0 m

Why the pruning differs is physical, not tuning. KL is hilly: unpruned, headwater ditches carry HAND_canal ≈ 0 onto the ridges and flood the high dry controls (Federal Hill, TTDI). An accumulation prune removes ditches that are not conveyance. Jakarta is a flat delta whose canals are *all* engineered conveyance — there are no hill headwater ditches to remove, and the gate is insensitive to pruning there. **Rule: prune in hilly cities; do not prune on deltas.**

The 4-city assessment — tested on all four before applying to any:

City	shipped pluvial-only HR	dry-control geometry	canal-HAND verdict
Kuala Lumpur	0.13 (under-powered)	all high ground (min 49.7 m)	applied
Jakarta	0.64	high ground (min 31.9 m)	applied — strict win, TSS improves
Singapore	0.71 (already strong)	moderate relief (min 6.7 m)	rejected — TSS 0.61 → 0.39-0.54 across every variant
Bangkok	0.25	degenerate (all controls < 5 m)	rejected — CRR 1.00 → 0.43, TSS 0.34 → 0.02

Stage-2 selection rule (the transferable finding). Retrofit canal-HAND **only** where (a) the shipped pluvial-only HR is demonstrably low **and** (b) the register's dry controls sit meaningfully **higher** than its positives — i.e. topographic position relative to drainage is informative. **Both are checkable before building anything.** The two rejections explain why: on a **flat pumped delta** (Bangkok) engineered drainage decouples flooding from topography, so no HAND variant can discriminate — it floods the low dry controls and specificity collapses; in an **already-skilful, heavily-drained city** (Singapore) it only adds false alarms.

Measured effect of the two applications:

	KL shipped → retrofit	Jakarta shipped → retrofit
pluvial extent (RP100/2020)	37.9 → 246 km ²	163 → 245 km ²
largest contiguous patch	0.18 → 6.5 km ²	—
documented flash-flood streets hit	13 % → 55 %	—
gate HR / CRR / TSS	0.65/1.00/0.65 → 0.71/0.92/0.63	0.76/1.00/0.76 → 0.85/0.92/0.77

KL trades a hill dry-control (CRR 1.00→0.92) for three recovered documented streets (Segambut Dalam, Jalan Tun Razak, Bangsar); TSS is **parity within the ±0.16 bootstrap CI** while HR rises 0.06 — the layer stops being near-redundant (pre-retrofit NO-PLUVIAL HR 0.58 vs ALL 0.65). Jakarta is a **strict win** — the headline TSS improves. Both scale monotonically with RP (KL ≈224→315 km², Jakarta 179→342 km² across RP10→RP1000/2100). (All extents in this subsection are as measured at the 2026-07-04 retrofit, before the 2026-07-07 bridge-closing scaled every city's pluvial extent ×1.20-1.28; the shipped present-day values are in §8.)

Shipped per-city configuration (post-retrofit — what the atlas and paper report)

City	Base solver (Stage 1)	Canal-HAND (Stage 2)	Shipped pluvial	Gate HR/CRR/TSS
Singapore	handfill (hybrid HAND, stage 1.5 m scaled , baseline 0.0696, cap 0.5 m)	rejected	base only	0.71 / 0.90 / 0.61
Kuala Lumpur	fillspill (cap 3.0 m)	applied (pruned ≥0.5 km ²)	max(base, canal-HAND)	0.71 / 0.92 / 0.63
Bangkok	fillspill (cap 3.0 m)	rejected	base only	0.34 / 1.00 / 0.34
Jakarta	handfill (stage 2.5 m scaled , baseline 0.1300, cap 1.0 m)	applied (unpruned)	max(base, canal-HAND)	0.85 / 0.92 / 0.77

("scaled" = the RP/climate stage-scaling of §6.3.2, stage × clip(excess/baseline, 0.4, 2.5); both handfill cities ship it, and the driver's Stage-1 lines carry the --pluvial-hand-stage-baseline flag so a clean rebuild reproduces it.)

Implementation note (STAGE 2 of the driver). Since 2026-07-13 the retrofit runs inside repro/run_atlas_fixed.sh as **STAGE 2**, after the Stage-1 solve loop. The order is load-bearing:

- 1 **canal-HAND applies** (scripts/_apply_kl_canal_pluvial.py, _apply_jakarta_canal_pluvial.py) — idempotent: each recomputes from the pristine *.tif.orig sidecar, so a resumed or repeated run is safe.
- 2 **bridge-closing** (_apply_city_pluvial_closing.py, all four cities) — backs the *current* raster up to *.preclose once and closes that. It must run **after** the applies, and must never read *.orig.
- 3 **summary/severity regen** (_regen_pluvial_summary_severity.py) — must run **last**, so the CSVs and severity rasters describe the final depths.

The regen is **read-only on the depth raster**. It formerly also re-masked each cell to the footprint stored in *.orig — a one-time repair for footprint damage the 2026-07-04 apply introduced, correct

only while the regen ran immediately after that apply. Once the closing exists it is actively destructive: *.orig predates the canal-HAND, whereas the closing deliberately keeps bridged cells falling *outside* that footprint, so re-masking silently deleted them (measured on jakarta_ssp585_2100_rp100: pluvial extent cut 337.2 → 334.4 km² at ≥ 0.10 m, −2.81 km², taking the cell off its published Table-II value of 337). Removed 2026-07-13: the depth the chain leaves is authoritative, and this step only derives the severity raster and the summary row from it.

Singapore's gate is new, not changed. The SG register (hotspots_expanded.csv, PUB *List of Flood-Prone Areas* Nov 2025 — 38 positives / 20 dry controls) was imported 2026-07-04; SG scores 0.71/0.90/0.61 **with no model change** — a validation result the model previously lacked, not a retrofit effect.

Runoff coefficient. Common to all four: --runoff-coeff-raster (WorldCover-derived) with scalar fallback --runoff-coeff (Singapore/KL 0.75, Jakarta 0.80).

7. Validation: model-blind hotspot gate

In plain terms. A model that is easy to rebuild can still be wrong, so this is the part where we try to catch it being wrong. For each city we wrote down two lists: places documented to have flooded, and places documented to have stayed dry. Then we checked whether the model floods the first list and leaves the second alone.

Two rules make it a real test rather than a formality. The lists are **frozen before** the final scoring, so the model cannot be quietly adjusted until it agrees. And nothing is reclassified once frozen — if the model floods a place on the frozen dry list, that counts against it permanently. (Two Jakarta labels were corrected on documented flood records before the final freeze; the published list records both, and Jakarta passes either way — see docs/presentation/SAFE2026-FAQ.md, F8.) Three scores come out: how many real flood spots it catches, how many dry spots it correctly leaves alone, and a single combined number where 0 is no skill and 1 is perfect.

Principle. Trust is established by a **pre-registered, model-blind location-skill gate** (scripts/validate_hotspots.py, math in scripts/hotspot_scoring.py), not by reproducibility alone. A frozen register of documented-**flooded** ("positive") and documented- ("control") localities is scored against the model's combined flood layer. The register is fixed before scoring and the model is never tuned against it.

7.1 The buffered hit test

Each register entry is a georeferenced point (lon, lat) with class ∈ {flood, dry}. A point scores a **hit** (sample_hit) iff any cell within a radius buffer exceeds the depth threshold:

```
hit(point) = TRUE    iff   ∃ cell ∈ window(point, r) : depth(cell) ≥ τ
τ = 0.10 m (depth threshold)    r = 50 m (hit radius)
```

The window is the pixel box ±[r/res] cells about the point (at 30 m, r = 50 m → ±2 cells), and the test is ≥ τ over finite cells in that box. A point outside the raster, or with no finite cells in its window, scores FALSE (a miss for a flood point, a correct rejection for a dry point). The buffer absorbs geocoding/DEM co-registration error; τ = 0.10 m is the nuisance-flood threshold.

7.2 Skill metrics (skill_scores)

With H = hits among P flood positives and F = false-alarms among N dry controls:

```
HR (hit rate / sensitivity)      = H / P
CRR (correct-reject rate / specificity) = (N - F) / N = 1 - F/N
TSS (True Skill / Peirce statistic) = HR + CRR - 1 ∈ [-1, 1]
```

TSS = 0 is no skill (random/flood-everything/dry-everything); TSS = 1 is perfect. TSS is used because it is **prevalence-independent** and rewards specificity equally to sensitivity — the **dry-control discipline**: inflating HR by flooding more area immediately raises F and crashes CRR, so the gate cannot be gamed by over-prediction. (Edge cases: empty positives → HR 0; empty negatives → CRR 1.)

7.3 Confidence interval — stratified bootstrap (bootstrap_tss_ci)

The TSS uncertainty is a **stratified percentile bootstrap** (not analytic). Positives and negatives are independent samples, so each of $n_{\text{boot}} = 10,000$ iterations resamples them **separately** with replacement and recomputes TSS:

```
for b in 1..10000:
  HR*_b = mean(resample(flood_hits, P))      # P draws with replacement
  CRR*_b = 1 - mean(resample(dry_hits, N))    # N draws with replacement
  TSS*_b = HR*_b + CRR*_b - 1
CI_95 = [ quantile(TSS*, 0.025) , quantile(TSS*, 0.975) ]    # percentile method
```

Deterministic (seed = 12345). **Significant discriminative skill $\Leftrightarrow$ the 95 % CI excludes zero.**

Reconciled (2026-06-22). *The implementation is a stratified percentile bootstrap (np.quantile on the resampled TSS) — there is no bias-correction z_0 or acceleration $\hat{a}$ term, so it is **not** BCa. An earlier paper draft called it "BCa"; the manuscript (safe2026-v5.tex) now reads "stratified percentile bootstrap", matching the code and this section.*

7.4 Gate criteria

Two gate forms exist in the codebase:

- **Absolute PASS gate** (evaluate_gate, the headline) — pre-registered and stricter:

PASS $\Leftrightarrow$ $HR \geq 0.70$ AND $CRR \geq 0.70$ ($--hr\text{-}floor$, $--crr\text{-}floor$)

- **Baseline-margin gate** (passes_numeric_gate, for "beats the open-baseline" claims) — $HR \geq floor$ AND TSS exceeds every baseline's TSS by a margin (default 0.20), tested with a **paired** stratified bootstrap (bootstrap_tss_diff_ci: resample one index set, apply to both classifiers, so shared sampling noise cancels) and a threshold-free **ROC-AUC** companion (roc_auc = Mann-Whitney $P(\text{score}_{\text{flood}} > \text{score}_{\text{dry}})$, with bootstrap_auc_diff_ci).

Present-day results (RP100, ≥ 0.10 m, 50 m radius, expanded/frozen registers — the **shipped post-retrofit atlas**, re-baselined 2026-07-12; Fisher = one-sided exact; reproduce with repro/reproduce_gate.sh):

City	pos/dry	HR	CRR	TSS [95% CI]	Fisher p	status
Jakarta	33/12	0.85	0.92	0.77 [0.54, 0.94]	4.5×10^{-6}	PASS
Kuala Lumpur	31/12	0.71	0.92	0.63 [0.38, 0.84]	2.6×10^{-4}	PASS (narrowest)
Singapore*	38/20	0.71	0.90	0.61 [0.41, 0.79]	7.9×10^{-6}	PASS
Bangkok†	32/7	0.34	1.00	0.34 [0.19, 0.50]	0.077	positive, not robustly significant (see †)

* Singapore = hybrid bare-earth + main-stem fluvial + handfill pluvial. † Bangkok's HR is bounded by the **out-of-domain** Chao Phraya catchment [R29] and sub-30 m street flooding. Its skill is **not robustly significant**: the frozen 7-control stratum is degenerate (7/7 correct $\rightarrow$ zero CRR variance, Fisher p = 0.077, marginal), and **expanding it to 10 documented dry controls (2026-07-18) flips the result to TSS 0.24 [−0.02, 0.47], Fisher p = 0.137 — CI crosses zero** (docs/runs/2026-07-18-bangkok-control-expansion.md). The flip turns on one borderline riverside control (Grand Palace, transient ankle water in 2011 held dry by pumping). This **corroborates the SAFE abstract's "positive but marginal in the fourth city"** — Bangkok is not counted among the three significant cities. All of Singapore / KL / Jakarta clear the stricter PASS bar and are significant by the exact test; Bangkok is the disclosed *structural* shortfall (§10), not a tuning failure. The **Stage-1 (pre-retrofit)** gate — Jakarta 0.76/1.00/0.76, KL 0.65/1.00/0.65 — lives in §6.3.5, where it is the evidence that selected each city's base solver.

8. Outputs & atlas structure

In plain terms. What you actually download is a folder per city, per climate scenario, per severity. Inside each are image-like map files (GeoTIFFs) that open in any GIS tool — one per hazard, holding the water depth in metres for every 30-metre square — plus a "severity" version that bins those depths into classes, and a small spreadsheet summarising how much area flooded and how deep it got.

One thing to know before comparing numbers: the summary spreadsheets count any cell with water at all, while the manuscript counts only cells at least 10 cm deep. The same map legitimately yields two different area figures. Neither is wrong — the depth files are the authoritative thing, and both numbers can be recomputed from them.

Per-cell output directory: `outputs/_fixed_atlas/<city>_<stem>_rp<rp>/` (Bangkok adds a `_polder` sibling holding the polder-dried composite that is the one scored/published). Inside, per hazard:

```
<cell>/<hazard>/rp_<rp>/<hazard>_depth_<scenario>_<horizon>_rp_<rp>.tif # depth
<cell>/<hazard>/rp_<rp>/<hazard>_severity_<...>.tif # classed severity
<cell>/summary_<label>_<horizon>.csv # per-hazard extent/stats
```

Derived layers.

- **Combined** = per-pixel **max** of coastal, fluvial, pluvial depth (computed for the maps in `_viz_*`; conceptually the screening layer).
- **Compound** = a joint-exceedance layer whose *footprint* is dependence-robust while its *depth* amplifies into the tail.

Headline extents (shipped post-retrofit atlas, measured from the rasters 2026-07-17).

Present-day RP100, per hazard, ≥ 0.10 m (the manuscript's reporting convention; the shipped `summary_*.csv` count at the engine's > 0 and therefore read higher — the convention note ships in every Zenodo README), km²:

City	Coastal	Fluvial	Pluvial
Singapore	0.9	28.5	111.2
Kuala Lumpur	— (inland)	70.2	296.2
Jakarta	182.1	180.9	297.6
Bangkok (polder composite)	68.0	349.1	229.1

The canonical headline (paper Table 2) is **SSP5-8.5/2100 RP100**: Singapore 2/37/133, KL 0/87/326, Bangkok 853/377/326, Jakarta 229/214/337 (coastal/fluvial/pluvial; verified raster-to-table 2026-07-13).

- **Bathtub bias** (full model domain, SSP5-8.5/2100, same basis as Table 2): the connectivity bathtub over-predicts the inertial coastal extent by **1.2×** (Jakarta, 229→283 km²) to **5.2×** (Bangkok, 853→4,432 km²). The inertial bars equal the Table 2 coastal extents exactly. The defended-coast ratio is effectively unbounded at present-day (the dike holds the design surge, so the inertial footprint approaches zero), so the bias is reported at the 2100 horizon where both layers are substantial (§6.1.3). Earlier drafts clipped this comparison to the city admin boundary (7.6×/1.6×), which did not reconcile with the full-domain Table 2 extents and was dropped.
- **KL pluvial** 0→41.7 km² (*Stage-1; the fill-spill fix — the "0" was the raingrid-drain bug*) → **296** km² shipped (canal-HAND retrofit + closing, §6.3.5).
- **Singapore** fluvial 180→**28.5** km² (main-stem) + pluvial 15→95 (*Stage-1 handfill*) → **111** shipped (stage scaling + closing).
- **Mitigation delta**: meeting SSP2-4.5 rather than SSP5-8.5 avoids ≈ 92 km² of Bangkok's 100-year coastal exposure by 2100.
- **Compound (Jakarta present RP100)**: joint footprint 502.1 vs marginal 501.9 km² (dependence-robust; independence adds ≤ 0.2 km² in every city — KL exactly 0); depth amplification +0.10 m (RP100) → +1.47 m (RP1000, 90th pct +3.26 m). Re-derived 2026-07-12 on the post-retrofit atlas (canal-HAND 07-04 + pluvial closing 07-07); the larger pluvial grew the marginal from 396.5 km² and diluted the RP1000 mean amp from +1.67 m (the earlier depth-capped value; +1.45 m pre-cap).

9. Reproducibility

End-to-end rebuild. `bash repro/run_atlas_fixed.sh` regenerates the full atlas (60 cells). It is **resumable**: any (city, stem, rp) whose summary CSV already exists is skipped, so an interrupted run continues where it stopped. It first regenerates the cheap derived rasters (sea-mask frame-fix, seawall DEMs, main-stem HANDs) idempotently, then runs the Stage-1 solve loop, then the **STAGE 2 post-solve chain** (canal-HAND applies → bridge-closing → summary/severity regen, in that load-bearing order — §6.3.5), so one command reproduces the shipped product.

Present-day RP100 fast slice. `bash repro/run_fixed_coastal_rp100.sh` produces just the present-day RP100 verification outputs (the visual/gate slice).

Interpreter. All runs use the Python 3.14 interpreter named in §2.

Representative invocation (Singapore, from the driver — flags verified against `scripts/run_multihazard.py --help`, 2026-06-21):

```
python scripts/run_multihazard.py --dem dem/_diag/singapore_seawall.tif \
--fluvial-hand-raster dem/_hand/singapore_hand_v3_mainstem.tif \
--hazard-levels data/singapore/hazard_levels_&lt;stem&gt;_rp&lt;rp&gt;.csv \
--scenario &lt;label&gt; --horizon &lt;hz&gt; --out-dir &lt;out&gt; \
--coastal-solver inertial --coastal-msl-egm2008 1.1588 \
--sea-mask-raster data/singapore/sea_mask_utm48n_framefix.tif \
--tidal-channel-raster data/singapore/river_mask_utm48n.tif --tidal-burn-elevation 2.0 \
--pluvial-model handfill --pluvial-hand-raster dem/_hand/singapore_hand_v3_hybrid.tif \
--pluvial-hand-stage 1.5 --pluvial-hand-stage-baseline 0.069602 --pluvial-depth-cap 0.5 \
--runoff-coeff-raster data/singapore/runoff_coeff_utm48n.tif --runoff-coeff 0.75 \
--fluvial-bankfull-rp 0
```

All flags above are confirmed present. `--clamp-negative-land` is a boolean-optional action, so `--no-clamp-negative-land` (used for Jakarta/Bangkok) is its negation.

Other notable flags (`run_multihazard.py --help`), for tuning the solvers:

- Coastal/inertial: `--coastal-seed-latlon`, `--seed-water-raster`, `--inertial-dt-max`, `--inertial-t-end`, `--inertial-convergence-tol`, `--manning-raster`, `--connectivity-neighbors`.
- Pluvial/rain: `--pluvial-dem-raster`, `--rain-storm-hours`, `--rain-total-hours`, `--drain-conveyance-m-s`, `--raingrid-workers`.
- Fluvial: `--major-river-raster`.
- Scope: `--only-hazard-types` (e.g. `fluvial,pluvial` for inland KL).

10. Limitations & known issues

In plain terms. *This is a screening tool. It is built to tell you which places and which scenarios deserve a closer look — not to size a flood wall, settle an insurance claim, or design a building. Three limits matter most for anyone using it without reading the rest of this document.*

It cannot see flooding driven from outside the mapped area, which is why Bangkok's score is capped: the river that flooded it in 2011 rises well upstream of the map's edge. It cannot see flooding smaller than 30 metres across, which is most street-corner flash flooding. And it reports a plausible severity, not a range — we have not carried the uncertainty in the input data through to the maps, so treat the depths as an upper-bound screen rather than a best estimate with error bars.

Everything below is reported rather than quietly fixed. Several items are things our own adversarial reviews found and we chose to disclose.

- Bangkok hit-rate is forcing-bounded, not terrain-bounded.** Accurate terrain cannot manufacture skill the forcing domain withholds: Bangkok's HR (0.34) is capped by the out-of-domain Chao Phraya mega-river [R29] and by sub-30 m street flooding. More accurate terrain helps a flood screen only *conditionally*.

- 1 **KL pluvial is resolution-bounded and looks spotty.** KL's flash floods are sub-grid; `fillspill` on steep terrain ponds in ~12,000 tiny disconnected pits. It is retained because it is gate-optimal (perfect CRR), but the map reads as speckle — an honest artifact of representing sub-grid flash floods at 30 m, not a model error.
- 1 **Handfill stage — gate-selected base; now standards-bracketed, with a measured held-out bound.** The Singapore (1.5 m) and Jakarta (2.5 m) *base* stages were chosen from the gate trade-off, which brushes against the "never tune to the gate" principle. Two mitigations executed 2026-07-18 (review C7 follow-ups):
- 2 **Bracketing:** external standards bracket both values (SG 1.5 $\in \approx[1.1, 2.0]$ m from PUB CoP drain geometry + freeboard + documented ponding; JKT 2.5 = top of the BNPB Jan-2020 0.2–2.5 m band) — the values are physically bounded, but the choice *within* the bracket saw the gate. Full derivation §6.3.2.
- 3 **Held-out event test (measured, logged, not re-tuned):** the 2024-10-15 DBKL flash-flood bulletin (10 roads absent from every register) scores **HR 0.30** at gate protocol (0.60 at road-scale 250 m radius) against the shipped layer — `docs/runs/2026-07-18-kl-held-out-event.md`. The register PASS is **register-class skill** (recurrent flood-prone localities); single-event road-bulletin generalisation is measured and bounded at 0.30, consistent with the sub-grid convective failure mode (item 2). Any per-event prediction claim must carry this number. Separately, the fixed stage made the layer **ignore the forcing** — resolved by the stage-scaling (§6.3.2), now passed directly by the driver's Stage-1 lines (`--pluvial-hand-stage-baseline`, 2026-07-17). [Residual: the within-bracket selection remains gate-visible — disclose in the paper's methods; the held-out number is the quantitative disclosure.]
- 1 **Uniform coastal crest is a simplification.** A single continuous crest per city is more conservative than documented per-segment crests; production should move to per-segment crests for realistic residual flooding behind defences.
- 1 **Jakarta coastal is real, not an artifact.** The persistent coastal depth is chronic below-sea-level subsidence; it should not be "corrected" away.

10.1 Adversarial methodology review — Round 3 (2026-07-09)

A third adversarial round ([full record](adversarial-review-2026-07-09.md)) challenged the rationale behind every major methodological choice and the cross-city heterogeneity, building on the two referee stress-tests (gate statistics; elevation-null and stage sensitivity). Outcomes:

- **Four paper claims corrected in** `ieee-kse2026.tex` — most materially, the mitigation-delta mechanism: on the bare-earth atlas the connectivity bathtub *saturates* the flat Bangkok delta under both 2100 pathways (avoided area ≈ 35 km² vs the inertial 92 km²), so it *washes out* the scenario signal rather than inflating it as the DSM-era-derived sentence claimed. Also fixed: the abstract's DSM-bias mechanism (the high bias **masks** documented flooding — measured bathtub extents are *larger* on bare earth — it does not drive over-prediction; the solver does), a stale pluvial-limitations sentence, and the KL pluvial union description.
- **Two "looks-wrong-but-isn't" audit traps documented:** `--fluvial-bankfull-rp 0` in the release script (bankfull subtraction is baked upstream in `fit_fluvial_glofas.py`; run-time 0 avoids double-subtraction), and Jakarta's pumped-polder rings existing in `apply_pumped_polder.py` but deliberately unapplied (Pluit/rob flooding is documented to defeat the pumps; Bangkok's 2011-proven pump core is the only drained polder — the asymmetry follows the evidence).
- **Heterogeneity register** (review §3): the per-city differences split into evidence-driven choices (terrain recipe, pump floor, canal pruning, IDF durations, depth caps — each with a documented anchor) and deliberately homogeneous simplifications (coastal Manning $n = 0.06$, the 3-1-2 h surge hydrograph) whose *uniformity* is the untested assumption.
- **Two OPEN sensitivity items** logged (not blocking): coastal $n \in \{0.03, 0.10\}$ and $t_{\text{end}} = 16$ h bracket for Bangkok 2100; held-out-event validation of the canal-HAND layer remains the standing pre-submission recommendation (with item 3 above, the base-stage re-anchoring).

10.2 Addendum — Round-3 follow-ups closed (2026-07-18)

The three ranked follow-ups from the Round-3 review are now executed ([full record](adversarial-review-2026-07-18-addendum.md); held-out run in [docs/runs/2026-07-18-kl-held-out-event.md](../runs/2026-07-18-kl-held-out-event.md)):

- **C6 (t_end):** measured — Bangkok-2100 inertial extent grows **+18 % at t_end = 16 h**, still not converged; 98.8 % of the added area is below the permanent SLR floor, so it is floor-fill propagation, not surge truncation. The 8 h extent is a lower bound; the durable (t_end-independent) bathtub over-prediction is the **above-floor 743 km²**, not the full 8 h ratio. Two paper-side framing decisions logged (do not change any shipped number).
- **C5 (Manning), regime split:** the friction lever is $\approx \pm 6$ % **undefended (Jakarta)** but $\approx \pm 25$ -**40 % on defended deltas (Bangkok)**: $n=0.03/0.06/0.10 \rightarrow 1300/920/710$ km²). The earlier "friction ~6 %" reassurance was undefended-only and is corrected.
- **C7 (held-out event):** the 2024-10-15 DBKL bulletin (10 held-out roads) scores **HR 0.30** at gate protocol — register skill is register-class, single-event generalisation is bounded (§10 item 3). Handfill base stages now standards-bracketed (§6.3.2).
- **C13 (Bangkok degenerate control stratum):** expanding the 7-control stratum to 10 documented dry controls flips Bangkok from bootstrap-significant to **TSS 0.24 [−0.02, 0.47], Fisher p 0.077→0.137** — confirming the significance was stratum-fragile, and corroborating the paper's "positive but marginal" framing. Not re-tuned; the paper's frozen register is untouched (docs/runs/2026-07-18-bangkok-control-expansion.md). §7.4 Bangkok row corrected to "not robustly significant".
- **C14 (KL IDF-duration sensitivity):** the KL gate is **byte-identical** across the design-depth bracket. Running the KL pluvial at RP100 excess 65 / 95 / 130 mm (spanning a 1-hour-duration design to the current) moves the base fill-spill extent only −22 % at the 1-hour end (29.4 vs 37.9 km²) and the shipped-dominant canal-HAND extent only **−5.4 %/+10 %** (246 / 260 / 287 km²) — because the canal-HAND stage $\text{clip}(0.5 \cdot \text{excess} / 0.095, 0.4, 2.5)$ floor-clips at 0.40 m at the 1-hour end. Scored at gate protocol, the 1-hour end (stage 0.40, pluvial 68.5 km²) and the current design (stage 0.50, 89.5 km²) give **identical HR 0.65 / CRR 0.92 / TSS 0.56** (no-closing basis; the closing lifts both equally to the shipped 0.71/0.92/0.63): the design-depth cut removes only the shallowest far-from-drain cells, which are neither documented flood streets nor dry controls. The IDF-duration choice is immaterial to KL's validation. (A definitive 1-hour refit still needs the MSMA station IDF coefficients, not in the repo; the bracket bounds the answer regardless.)

(See also docs/limitations_register.md.)

11. Appendices

11.1 Per-city configuration (from repro/run_atlas_fixed.sh)

Param	Singapore	Jakarta	Bangkok	Kuala Lumpur
Flood DEM	_diag/singapore_seawall	_diag/jakarta_seawall	_diag/bangkok_seawall	kl/dem_bareearth_kl_eth_present_conditioned
Fluvial HAND	singapore_hand_v3_mainstem	jakarta_hand_v3_mainstem	hand_trunk_v3_debiased	kl_hand_mainstem_v3eth
Coastal solver	inertial	inertial	inertial	— (inland)
MSL-EGM2008	1.1588	0.9976	1.1785	—
Sea-mask	sea_mask_utm48n_framefix	sea_mask_utm48s_framefix	sea_mask_utm47n_framefix	sea_mask_utm47n
Pluvial model	handfill	handfill	fillspill	fillspill

Param	Singapore	Jakarta	Bangkok	Kuala Lumpur
Pluvial HAND / stage / cap	hybrid / 1.5 / 0.5	jakarta_hand_v3 / 2.5 / 1.0	— / — / 3.0	— / — / 3.0
Handfill stage baseline (§6.3.2)	0.069602	0.129999	—	—
Runoff coeff	0.75	0.80	raster	0.75
Coastal crest (m MSL)	+3.0	+2.0	+2.10	—
Polder post-process	—	—	yes	—
Hazards	coastal+fluvial+pluvial	coastal+fluvial+pluvial	coastal+fluvial+pluvial	fluvial+pluvial

11.2 Glossary (plain-language)

Everyday definitions of the terms and symbols used above, grouped by topic. Where a term also has a precise technical meaning, the section that defines it rigorously is noted.

The three flood types ("hazards").

- **Coastal flooding** — the sea coming inland: high tide, plus a storm surge, plus long-term sea-level rise. (§6.1)
- **Fluvial flooding** — a river overflowing its banks. (§6.2)
- **Pluvial flooding** — rain falling faster than the ground and drains can carry it away, so it ponds in low spots or backs up beside drains. This is the "flash-flood" type that global products usually leave out, and it is the dominant hazard in three of our four cities. (§6.3)

Terrain (the ground the water flows over).

- **DEM** (Digital Elevation Model) — a grid of ground-height numbers, one per 30 m square. The single most important input, because water runs downhill.
- **DSM vs bare-earth DEM** — the raw satellite surface (**DSM**) includes the tops of buildings and trees; the **bare-earth** version strips those off to leave the true ground. We build our own bare-earth surface (§4) because buildings and trees would otherwise look like high, dry land and hide real floods.
- **Subsidence** — land slowly sinking, here mostly from groundwater pumping. North Jakarta sinks up to ~25 cm per year (locally 20–28), which matters as much as sea-level rise — a decade of it outweighs a century of AR6 sea-level rise at that spot. Bangkok now sinks 1–2 cm per year, having historically reached ~12 cm per year before pumping controls. Only Jakarta and Bangkok get a subsidence correction, because only they have documented InSAR rates; Singapore and Kuala Lumpur have no comparable signal. (§5.4, §11.1)
- **HAND** (Height Above Nearest Drainage) — how high a spot sits above the nearest river or drain it flows to: 0 m means you are in the channel, a large value means you are safely above it. A natural "how flood-prone" ruler. *Main-stem* HAND measures height above the big river trunks only; *dense* HAND measures height above every small drain too. (§6.2)

How the water is simulated ("solvers").

- **Bathtub vs inertial** — two ways to spread the water. **Bathtub** instantly fills every connected low spot up to the water level, like a filling bath — fast, but it over-floods, because real water needs time and meets friction as it travels. **Inertial** (local-inertia shallow-water, Bates 2010) moves the water step by step with momentum and friction, so far-inland low ground the water cannot physically reach within a ~6-hour surge stays dry. We use inertial for the coast; the two can differ ~5x on a defended delta. (§6.1)
- **fill-spill vs handfill** — two ways to model rain ponding. **fill-spill** collects rain in closed basins and spills the overflow to the next basin downhill (suits flat, pit-dominated Bangkok). **handfill** floods the low land beside every drain (suits canal-dense cities where rain backs up along drains). Terrain sets the choice, and using the wrong one measurably lowers the trust score. (§6.3)

- **Manning's n** — one number for how rough the ground is, i.e. how much it slows moving water. Smooth pavement ~0.03, vegetated floodplain ~0.10; we use 0.06. Rougher ground = water spreads less far. (§6.1, §11.4)
- **Surge hydrograph** — the shape of a storm surge over time: how fast it rises, how long it holds at its peak, how it falls back. We use a generic 3 h-rise / 1 h-hold / 2 h-fall shape. (§6.1.2)
- **CFL / timestep** — a numerical safety rule: the simulation may only advance time in steps small enough that water never jumps more than one cell at once, or the maths becomes unstable. CFL_ALPHA = 0.7 is that safety margin. (§6.1.1, §11.4)

Design events and forcing (what drives the flood).

- **Return period (RP)** — how rare an event is. "RP100" (100-year) means a severity with a 1-in-100 chance of being exceeded in *any given year* — not "once every 100 years." We map RP10, RP100, and RP1000.
- **MSL / WSE / EGM2008** — **MSL** is mean sea level (the average the tide swings around); **WSE** is water-surface elevation (the actual height of the water surface at a moment); **EGM2008** is the global height reference (the "zero") every elevation is measured from.
- **GEV / Gumbel** — standard statistical curves describing how extremes (the biggest surge or heaviest rain of the year) get rarer as they get larger. We fit them to observed records to read off the RP100 severity. (§5.1, §5.3)
- **IDF** (Intensity-Duration-Frequency) — a country's official design-rainfall standard: how much rain falls in a given time span at a given return period. Anchoring to each national IDF closes a 28-62% gap that global rainfall data carries in this region. (§5.3)
- **Runoff coefficient** — the fraction of rain that runs off rather than soaking in. Paved city ~0.75-0.80; it scales rainfall into flood volume. (§5.3, §6.0)

The trust test (validation).

- **Hotspot register** — a frozen list of real places documented to flood (*positives*) and documented to stay dry (*controls*). Fixed before scoring, so the model cannot be quietly tuned to match it. (§7)
- **HR / CRR / TSS** — the three trust scores. **HR** (hit rate) = fraction of known flood spots the model floods (sensitivity). **CRR** (correct-reject rate) = fraction of known dry spots the model correctly leaves dry (specificity). **TSS** (true-skill statistic) = HR + CRR - 1, one number from -1 to 1 where 0 is no skill and 1 is perfect. (§7.2)
- **Significant vs PASS** — **significant** is the weaker bar (the model tells flood from dry better than chance); **PASS** is the stricter, pre-registered bar (HR and CRR both at least 0.70). Three of the four cities PASS. (§7.4)
- **Expanded register** — the larger frozen hotspot list behind the headline scores (vs a smaller base list).

Other.

- **Polder** — a low area kept dry by embankments and pumps. Bangkok's pumped city-centre polder is modelled as staying dry; Jakarta's flooding polders are not. (§6.1.5)
- **Frame-fix** — a fix that removes a data-border artefact from the sea-mask so the coastal solver does not flood inward from the edge of the map. (§4)

11.3 Script index (descriptions verified from each file's header, 2026-06-21)

- scripts/run_multihazard.py — the three-hazard solver + CLI.
- scripts/validate_hotspots.py — "Documented-hotspot validation — the primary gate (Plan 2)", generalised to a --ci bootstrap confidence interval.

- `scripts/_fix_coastal_seawall.py` — "Fix coastal over-prediction: burn a continuous coastline seawall crest into" the DEM.
- `scripts/_fix_sea_mask_frame.py` — "Fix spurious coastal edge-flooding: the `build_sea_mask` NaN-BFS sweeps the" nodata border into sea; strip it.
- `scripts/_fix_sg_mainstem_hand.py` — "Fix SG fluvial over-prediction: rebuild the Singapore HAND against a" main-stem drainage network.
- `scripts/_fix_jakarta_mainstem_hand.py` — "Jakarta main-stem HAND: rebuild HAND against a main-stem drainage network".
- `scripts/apply_pumped_polder.py` — "Apply a documented pumped-polder drainage floor to a present-day flood composite" (Bangkok).
- `scripts/_viz_make_maps_fixed.py` — "Leaflet flood-map of the FIXED present-day verification run (outputs/_fixed)".
- `repro/run_atlas_fixed.sh`, `repro/run_fixed_coastal_rp100.sh` — drivers.

11.4 Parameter register (every computing step)

Every numerical parameter in the model, grouped by computing step, with value, rationale, and reference. DEM and forcing parameters are tabulated in §4.1 and §5.1–5.4; this register covers the solver and gate constants. Source of truth: the module named in each section.

How to read this register (plain-language). A "parameter" is a single number the model uses — think of it as one setting on a dial. Each row names the dial, gives the value we set it to, says why, and — in the last column — labels **where that number comes from** so you can judge how much to trust it. The label matters more than it looks: it separates numbers pinned to outside evidence (published science, official design standards, or values we measured from data) from numbers that are our own engineering judgement with no external source. One label, **GATE-SELECTED**, marks the honest weak spot — a value we chose while we could see how it moved the trust score. The whole point of labelling every number this way is that a reader can see, at a glance, which settings are anchored and which are judgement calls, rather than having to take the model on faith.

Provenance classes. Every parameter is explicitly classed, so that a reader can tell what is anchored to external evidence from what is an engineering choice. A value with no literature source says so — that is the honest statement of a screening assumption, and it is what makes the assumption auditable rather than hidden:

Class	In plain terms — how much is it anchored?	Meaning	Example
LITERATURE	Backed by published science	value (or its range) is taken from published, citable science	Manning $n = 0.06 \leftarrow$ Chow 1959 / Arcement & Schneider 1989
STANDARD	Backed by an official design standard	value comes from a published national/engineering design standard	IDF anchors $\leftarrow$ PUB / JPS-MSMA / TMD-RID / BMKG; defence crests
MEASURED	We derived it from real data (not the trust test)	value is fitted/derived from observed data <i>in this work</i> , never from the gate	(b_0, f) $\leftarrow$ held-out ICESat-2/GEDI; MSL offsets; GEV fits
PHYSICAL/NUMERICAL	A law of physics or a computing requirement	a physical constant or a scheme-stability requirement	g ; CFL $\alpha = 0.7 \leftarrow$ Bates 2010
SCREENING CHOICE	Our judgement call, reasoned but unsourced	engineering judgment with a stated rationale but no literature source ; the residual assumption class	depression filters; surge-hydrograph shape; velocity-head margin
GATE-SELECTED	Judgement call that saw the trust score — the disclosed weak spot	chosen with validation-gate visibility — a disclosed weakness , flagged for re-anchoring	handfill stage (§10 item 3)

Where a SCREENING CHOICE has been sensitivity-tested, the measured range is given in its row; where it has not, the row says so (these are the open items in §10.1's review).

Coastal — inertial solver (model/inertial_wave_model.py, §6.1)

In plain terms: the settings for the coastal flood simulation — how the water flows (gravity, roughness), how long the storm surge lasts, and the numerical safety limits that keep the simulation stable. Most are physics or computing requirements, not free choices.

Param	Value	Rationale	Reference
g	9.806 m s ⁻²	standard gravity	PHYSICAL — constant
CFL_ALPHA	0.7	explicit-scheme Courant safety factor (<1); $\Delta t \leq \alpha \cdot \Delta x / (\sqrt{g d} + u)$	NUMERICAL — Bates 2010 [R1] (stability requirement of the explicit scheme, not a free knob)
MIN_DEPTH	1×10 ⁻³ m	dry-interface threshold (no flux below)	NUMERICAL — wetting/drying floor; 3 orders below the 0.10 m reporting threshold so it cannot affect extent
n (Manning)	0.06 s·m ^(-1/3) , spatially uniform	urban/coastal floodplain roughness; developed-floodplain tables give 0.05–0.15, so 0.06 is a low-mid value	LITERATURE — Chow 1959 [R7] (standard n tables); Arcement & Schneider 1989 [R8] (USGS WSP 2339 floodplain-roughness guide). <i>Not</i> Bates 2010, which sources the solver, not the roughness. Sensitivity measured, regime-dependent. Undefended (C5, repro/coastal_manning_sensitivity.sh): n=0.06/0.08/0.10 → Jakarta present-RP100 182.1/177.1/171.1 km ² (–6.1 % at 0.10), IoU 0.94. Defended (2026-07-18, repro/coastal_c6_bracket.sh): n=0.03/0.06/0.10 → Bangkok-2100 1300.0/919.9/710.2 km ² (+41 %/–23 %) — behind a dike the extent is the overtopping run-out distance, so friction is first-order. The lever is ≈±6 % undefended but ≈±25–40 % on defended deltas; the shipped Bangkok cell uses the polder (removes the sensitive interior), and even n=0.03 stays ~3.4× below the bathtub. See the 2026-07-18 addendum
dt_max	30 s	timestep ceiling; the adaptive CFL test reduces it further	NUMERICAL — ceiling only; CFL binds in practice
t_end	8 h (–inertial-t-end)	must exceed the 6 h surge window so the peak is captured and the surface settles	SCREENING CHOICE — set from the hydrograph length (below). Sensitivity measured 2026-07-18 (C6 closed, repro/coastal_c6_bracket.sh): Bangkok-2100 extent grows +18.1% at t_end=16 h and is still not converged; 98.8% of the added area is below the permanent SLR floor, so this is SLR-floor propagation (the 8 h extent is a lower bound on the floor equilibrium), not surge-peak truncation. See the 2026-07-18 addendum
convergence_tol	1×10 ⁻³ m	early-stop on mean $ \Delta d $; 100× below the 0.10 m reporting threshold	NUMERICAL — cannot move reported extent
convergence_window	100 (disabled when SLR floor > 0)	forces the full surge simulation whenever a permanent floor exists	NUMERICAL — correctness guard, not a tuning knob
surge hydrograph	3 h ramp / 1 h hold / 2 h recession, on the MSL+SLR floor	asymmetric ramp avoids dry-bed shock; recession stops at the floor to preserve the permanent SLR signal	SCREENING CHOICE — synthetic shape, no literature source. A real surge hydrograph is site- and storm-specific; this is a generic design envelope. The <i>peak</i> (which sets extent) is forced by water_level_m; the shape affects only how long the peak is held. <i>Duration sensitivity untested</i> (review item C6, §10.1)
physical cap	(peak_WSE – bed) + 0.2 m	strips inlet wave-amplification; $0.2 \approx \frac{1}{2}v^2/g$ at $v = 2 \text{ m s}^{-1}$	SCREENING CHOICE (velocity-head margin from first principles). Fixed 2026-06-24: was $\max(0, \text{bed})$, which under-capped below-MSL depth on subsided coasts; Jakarta coastal re-solved (§6.1.1)
crop pad	32 cells	solver-bbox buffer	performance

Fluvial — HAND (flood_depth_model.flood_depth_hand, §6.2)

In plain terms: how the river-flood layer decides which land counts as "near the river." The accumulation threshold sets how big a channel must be to count as a real river trunk (rather than a small drain), so the flood is tied to the main river, not to every ditch.

Param	Value	Rationale	Reference
accumulation threshold	≥50 km ² (SG, JKT); ≥180 km ² (KL, Klang trunk)	main-stem trunk, not dense canals	HAND [R2,R3]; this work
--fluvial-bankfull-rp	0 (production)	full design stage mapped via HAND (no overbank subtraction)	this work
channel mask	channel cells → 0	conveyance ≠ inundation	this work

Pluvial — fill-spill cascade (model/pluvial_model.py, §6.3.1)

In plain terms: the rain-ponding model's rules for what counts as a real basin that collects water — deep enough to not be noise, shallow/small enough to not be a quarry or lake — and how deep ponding is allowed to get. These are judgement calls, each with a stated reason.

Param	Value	Rationale	Reference
WET_THRESHOLD_M	0.05 m	cells shallower are set dry	SCREENING CHOICE — below the 0.10 m reporting threshold, so it cannot move reported extent
MIN_DEPRESSION_DEPTH_M	0.5 m	reject depressions shallower than the DEM's own noise as spurious	SCREENING CHOICE — no literature source. Rationale: GLO-30's local (cell-to-cell) vertical noise makes sub-0.5 m "basins" indistinguishable from DSM artefact. <i>Swept 0.4–0.8 m in the v5 country work: every level that de-speckles loses a documented hotspot somewhere — no threshold separates noise from signal, so the data raster is left pristine and speckle is handled at display level</i>
MAX_DEPRESSION_DEPTH_M	3.0 m	exclude quarries/valleys/reservoirs — not urban rain ponding	SCREENING CHOICE — no literature source. Rationale: a >3 m closed basin is a landform or waterbody, not a street-scale ponding site; excluded like the sea
MIN_DEPRESSION_AREA_CELLS	9 (≈0.8 ha)	drop sub-pixel inter-building voids	SCREENING CHOICE — resolvability argument: <9 cells at 30 m is at/below the DSM's ability to define a basin
depth cap	3.0 m (BKK, KL)	upper bound on ponded depth	SCREENING CHOICE — no literature source. A life-safety/plausibility bound on a steady-state model that has no outlet timing; binds rarely
depression fill	pysheds (priority-flood family)	hydrological conditioning	Wang & Liu 2006 [R19] / pysheds [R27]; cascade after Barnes 2021 [R4] (<i>paper cites Planchon-Darboux [R18] — reconcile</i>)

Pluvial — handfill (flood_depth_hand on dense HAND, §6.3.2)

In plain terms: the rain-flood model for canal-dense cities — how high rain pools beside the drains (the "stage"), how that pooling scales up with bigger storms, and a cap on how deep it can get. The stage is the one setting that saw the trust score, so it is flagged and standards-bracketed rather than presented as externally fixed.

Param	Value	Rationale	Reference
--pluvial-hand-stage	1.5 m (SG), 2.5 m (JKT)	rain-pool extent above drainage (base, present-RP100)	GATE-SELECTED within a standards-bracketed range (2026-07-18, §6.3.2): SG 1.5 ∈ ≈[1.1, 2.0] m (PUB CoP drain depths + 15 % freeboard + documented 0.07–0.76 m ponding); JKT 2.5 = top of BNPB Jan-2020 0.2–2.5 m depth band. Held-out single-event HR 0.30 (§10 item 3)
--pluvial-hand-stage-baseline	0.0696 (SG), 0.1300 (JKT)	present-RP100 excess; enables RP/climate stage scaling stage ×clip(level/baseline,0.4,2.5)	this work, 2026-06-22 (§6.3.2)
--pluvial-depth-cap	0.5 m (SG), 1.0 m (JKT)	documented flash-flood depth bound	flash-flood records
near-channel mask	HAND < 0.05 m, + river_mask → 0	channels = conveyance	this work

Pluvial — raingrid (alternative, model/pluvial_rain_model.py, §6.3.3)

In plain terms: an alternative, more physical rain model (not used in the shipped atlas) — how long the storm runs and how rough the ground is. Listed for completeness.

Param	Value	Rationale	Reference
storm_duration	3600 s (1 h)	matches 1 h IDF parameterisation	[R23]
total_duration	5400 s (1.5 h)	storm + settling to capture peak	this work
Manning n	clip(0.11 – 0.08·C, 0.03, 0.10)	land-cover-derived roughness	this work
denoise min_cluster	6 cells (≈0.5 ha)	drop ponding speckle	this work
depth floor	0.05 m	strip sub-threshold sheet	this work

Validation gate (scripts/hotspot_scoring.py, §7)

In plain terms: the scoring rules for the trust test — how deep water must be to count as "flooded" (0.10 m), how close a prediction must land to a real flood site to count as a hit (50 m), and the pass mark. All fixed before scoring, which is what makes the test honest.

Param	Value	Rationale	Reference
depth threshold τ	0.10 m	a cell counts as "flooded" at ≥ 0.10 m	SCREENING CHOICE — no literature source. Rationale: below ~0.1 m the depth is within model/DEM noise and is not damaging; 0.1 m is also the conventional minimum mapped depth in global flood-hazard products. Pre-registered before scoring and applied identically to every city/RP
hit radius r	50 m $\rightarrow \pm 2$ px square window at 30 m	absorbs geocoding + co-registration error (§7.1)	SCREENING CHOICE , anchored to an external quantity: Nominatim geocoding precision (~one city block). Fixed before scoring and never varied to move a verdict — this is the discipline that matters more than the value
n_boot	10,000	stable percentile CI	LITERATURE — Efron & Tibshirani [R25]
seed	12345	deterministic reproducibility	NUMERICAL — any fixed seed; stated so results are bit-reproducible
PASS	HR ≥ 0.70 $\wedge$ CRR ≥ 0.70 $\wedge$ TSS CI > 0	pre-registered joint sensitivity/specificity bar	SCREENING CHOICE (pre-registered) — 0.70 is a conventional "useful skill" bar, not a derived constant; fixed <i>before</i> scoring, which is what makes it a gate rather than a fit. TSS = Peirce [R24]. Exact one-sided Fisher p reported alongside (the bootstrap flatters an all-correct control stratum)
baseline margin	0.20	"beats open baseline" gate	SCREENING CHOICE — pre-registered margin

12. References

Most entries match the manuscript bibliography (docs/paper/safe2026-v5.tex). Entries marked **[verify]** are tool/product citations I did not confirm to an exact reference.

- **[R1]** Bates, Horritt & Fewtrell (2010), *A simple inertial formulation of the shallow water equations*, J. Hydrol. 387:33–45.
- **[R2]** Nobre et al. (2011), *Height Above the Nearest Drainage*, J. Hydrol. 404:13–29.
- **[R3]** Rennó et al. (2008), *HAND, a new terrain descriptor... SRTM-DEM*, Remote Sens. Environ. 112:3469–3481.
- **[R4]** Barnes, Callaghan & Wickert (2021), *Computing water flow through complex landscapes — Part 3: Fill-Spill-Merge*, Earth Surf. Dynam. 9:105–121.
- **[R5]** Lenderink et al. (2017), *Super-Clausius-Clapeyron scaling of extreme hourly convective precipitation*, J. Climate 30:6037–6052.
- **[R6]** IPCC AR6 WGI / Fox-Kemper et al. (2021), *Ocean, Cryosphere and Sea Level Change*.
- **[R7]** Chow, V.T. (1959), *Open-Channel Hydraulics*, McGraw-Hill — standard Manning's n tables; developed/urban overbank n ≈ 0.05 –0.15.
- **[R8]** Arcement & Schneider (1989), *Guide for Selecting Manning's Roughness Coefficients for Natural Channels and Flood Plains*, USGS Water-Supply Paper 2339 — the standard floodplain-roughness selection guide.

- **[R9]** Muis et al. (2016), *A global reanalysis of storm surges and extreme sea levels*, Nat. Commun. 7:11969.
- **[R10]** Muis et al. (2020), *A high-resolution global dataset of extreme sea levels...*, Front. Mar. Sci. 7:263.
- **[R11]** Hawker et al. (2022), *A 30 m global map of elevation with forests and buildings removed (FABDEM)*, Environ. Res. Lett. 17:024016.
- **[R12]** Lang et al. (2023), *A high-resolution canopy height model of the Earth*, Nat. Ecol. Evol. 7:1778–1789.
- **[R13]** Sirko et al. (2021), *Continental-scale building detection... (Google Open Buildings)*, arXiv:2107.12283.
- **[R14]** Neuenschwander & Pitts (2019), *The ATL08 land and vegetation product for ICESat-2*, Remote Sens. Environ. 221:247–259.
- **[R15]** Dubayah et al. (2020), *The Global Ecosystem Dynamics Investigation (GEDI)*, Sci. Remote Sens. 1:100002.
- **[R16]** Zanaga et al. (2022), *ESA WorldCover 10 m 2021 v200*, Zenodo doi:10.5281/zenodo.7254221.
- **[R17]** ESA & Airbus (2022), *Copernicus DEM GLO-30*, doi:10.5270/ESA-c5d3d65.
- **[R18]** Planchon & Darboux (2002), *A fast, simple and versatile algorithm to fill the depressions of DEMs*, Catena 46:159–176.
- **[R19]** Wang & Liu (2006), *An efficient method for identifying and filling surface depressions (priority-flood family)* — **[verify: confirm pysheds' exact fill algorithm]**.
- **[R20]** Chaussard et al. (2013), *Sinking cities in Indonesia*, Remote Sens. Environ. 128:150–161.
- **[R21]** Abidin et al. (2011), *Land subsidence of Jakarta...*, Nat. Hazards 59:1753–1771.
- **[R22]** Phien-wej, Giao & Nutalaya (2006), *Land subsidence in Bangkok, Thailand*, Eng. Geol. 82:187–201.
- **[R23]** National IDF / drainage standards: PUB (Singapore), JPS-MSMA (Malaysia), TMD-RID (Thailand), BMKG (Indonesia).
- **[R24]** Peirce (1884), *The numerical measure of the success of predictions*, Science 4:453–454 (true-skill / Peirce skill statistic).
- **[R25]** Efron & Tibshirani (1993), *An Introduction to the Bootstrap*, Chapman & Hall.
- **[R26]** Manning (1891), *On the flow of water in open channels and pipes*, Trans. ICE Ireland.
- **[R27]** pysheds (M. Bartos), simplified flow-modelling library — **[verify exact citation/version]**.
- **[R28]** Pronk, Hooijer, Eilander et al. (2024), *DeltaDTM: A global coastal digital terrain model*, Sci. Data 11:273, doi:10.1038/s41597-024-03091-9 (Singapore hybrid low terrain).
- **[R29]** Komori et al. (2012), *Characteristics of the 2011 Chao Phraya River flood*, Hydrol. Res. Lett. 6:41–46.
- **[R30]** Alfieri et al. (2018), *A global network for operational flood risk reduction (GloFAS)*, Environ. Sci. Policy 84:149–158.

(Numbering note, 2026-07-17: [R7]/[R8] were each accidentally assigned twice; Alfieri moved to [R30] and the orphaned T_TIDE entry (Pawlowicz 2002) was removed — the coastal forcing is de-meanned annual-maximum total water level, never harmonically de-tided, so T_TIDE is not used anywhere in the pipeline.)